

B1 Assignment : Vector-3D

SINGLE CORRECT TYPE QUESTIONS

V01/1/001. If a point P whose position vector is $\vec{a}$, not lying on the plane ABC where A(2, 3, 0) B(3, -2, 0) C(2, -1, 0). Then distance of P from plane ABC is given by :

- (A) $|\vec{a} \cdot \hat{k}|$ (B) $\left| (\vec{a} - (2\hat{i} + 3\hat{j})) \cdot (\hat{i} - 5\hat{j}) \right|$
 (C) $\left| (\vec{a} - (2\hat{i} + 3\hat{j})) \cdot (\hat{j}) \right|$ (D) $\left| (\vec{a} \cdot (\hat{i} - 5\hat{j})) \right|$

Ans. A

Sol. A, B, C are lying on x-y plane

So distance of P($\vec{a}$) from x-y plane = $|\vec{a} \cdot \hat{k}|$

V01/1/002. If $\vec{a}$ and $\vec{b}$ are non - zero vectors which are linearly dependent such that $\frac{|\vec{a} + \vec{b}|}{|\vec{a} - \vec{b}|} = 2, |\vec{b}| > |\vec{a}|$ then :

- (A) $\vec{b} = 3\vec{a}$ (B) $\vec{b} = -3\vec{a}$
 (C) $\vec{b} = 2\vec{a}$ (D) $\vec{b} = -2\vec{a}$

Ans. A

Sol. Let $\vec{b} = \lambda \vec{a}$

$$\Rightarrow \frac{|\vec{a} + \lambda \vec{a}|}{|\vec{a} - \lambda \vec{a}|} = 2 \Rightarrow \frac{1 + \lambda}{1 - \lambda} = \pm 2 \Rightarrow \lambda = 3, \frac{1}{3} \Rightarrow \vec{b} = \begin{cases} 3\vec{a} \\ \frac{\vec{a}}{3} \end{cases}$$

$$\Rightarrow \vec{b} = 3\vec{a} \quad (\because |\vec{b}| > |\vec{a}|)$$

V01/1/003. If $\vec{a}$ and $\vec{b}$ are perpendicular unit vectors and vector $\vec{c}$ is such that $\vec{c} = \vec{a} + \vec{b}$, then $(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) + (\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) + (\vec{c} \times \vec{a}) \cdot (\vec{a} \times \vec{b})$ is :

- (A) 0 (B) 1 (C) -1 (D) $\vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}$

Ans. C

Sol. $(\vec{a} \cdot \vec{b})(\vec{b} \cdot \vec{c}) - (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{b}) + (\vec{b} \cdot \vec{c})(\vec{c} \cdot \vec{a})$

$$- (\vec{b} \cdot \vec{a})(\vec{c} \cdot \vec{c}) + (\vec{c} \cdot \vec{a})(\vec{a} \cdot \vec{b}) - (\vec{c} \cdot \vec{b})(\vec{a} \cdot \vec{a})$$

$$= 0 - (\vec{a} \cdot \vec{c}) + (\vec{b} \cdot \vec{c})(\vec{c} \cdot \vec{a}) - 0 + 0 - (\vec{c} \cdot \vec{b}) \quad (\because \vec{a} \cdot \vec{b} = 0)$$

$$\begin{aligned}
&= (\vec{b} \cdot \vec{c})(\vec{c} \cdot \vec{a}) - (\vec{c} \cdot \vec{a}) - (\vec{b} \cdot \vec{c}) + 1 - 1 \\
&= ((\vec{b} \cdot \vec{c}) - 1)((\vec{c} \cdot \vec{a}) - 1) - 1 \\
&= (1 - 1)(1 - 1) - 1 = -1 \quad (\because \vec{c} = \vec{a} + \vec{b})
\end{aligned}$$

V01/1/004. Consider three vectors $\vec{v}_1, \vec{v}_2$ and $\vec{v}_3$ such that $\vec{v}_1 = \vec{v}_2 - \vec{v}_3$ where $\vec{v}_1 = (\vec{a} \times \hat{i}) \times \hat{i}$, $\vec{v}_2 = (\vec{a} \times \hat{j}) \times \hat{j}$ and $\vec{v}_3 = (\vec{a} \times \hat{k}) \times \hat{k}$:

- (A) $\vec{a} \cdot \hat{j} = 0$ (B) $\vec{a} \cdot \hat{i} = 0$
(C) $\vec{a} \cdot \hat{k} = 0$ (D) $\vec{v}_1 \cdot \vec{v}_2 = (\vec{a} \cdot \hat{j})^2$

Ans. A

Sol.

$$\begin{aligned}
\vec{v}_1 &= (\vec{a} \times \hat{i}) \times \hat{i} = (\vec{a} \cdot \hat{i}) \hat{i} - \vec{a} \\
\vec{v}_2 &= (\vec{a} \times \hat{j}) \times \hat{j} = (\vec{a} \cdot \hat{j}) \hat{j} - \vec{a} \\
\vec{v}_3 &= (\vec{a} \times \hat{k}) \times \hat{k} = (\vec{a} \cdot \hat{k}) \hat{k} - \vec{a} \\
\vec{v}_1 &= \vec{v}_2 - \vec{v}_3 \\
\Rightarrow (\vec{a} \cdot \hat{i}) \hat{i} - \vec{a} &= (\vec{a} \cdot \hat{j}) \hat{j} - \vec{a} - (\vec{a} \cdot \hat{k}) \hat{k} + \vec{a} \\
\Rightarrow (\vec{a} \cdot \hat{i}) \hat{i} - \{(\vec{a} \cdot \hat{i}) \hat{i} + (\vec{a} \cdot \hat{j}) \hat{j} + (\vec{a} \cdot \hat{k}) \hat{k}\} &= (\vec{a} \cdot \hat{j}) \hat{j} - (\vec{a} \cdot \hat{k}) \hat{k} \\
&= 2(\vec{a} \cdot \hat{j}) \hat{j} = 0 \Rightarrow \vec{a} \cdot \hat{j} = 0
\end{aligned}$$

V01/1/005. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ are unit vectors inclined to each other at an angle of $\frac{2\pi}{3}$. If $\hat{p}, \hat{q}$ and $\hat{r}$ are unit vectors along the angle bisectors of $\vec{a} \wedge \vec{b}, \vec{b} \wedge \vec{c}$ & $\vec{c} \wedge \vec{a}$ respectively, then $[\hat{p} \quad \hat{q} \quad \hat{r}]$ is equal to :

- (A) 1 (B) 0 (C) $\frac{1}{\sqrt{3}}$ (D) $\sqrt{3}$

Ans. B

Sol.

$$\vec{a} \wedge \vec{b} = \vec{b} \wedge \vec{c} = \vec{c} \wedge \vec{a} = \frac{2\pi}{3}$$

$$\therefore \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix} = [\vec{a} \ \vec{b} \ \vec{c}]^2$$

$$\Rightarrow [\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & 1 & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & 1 \end{vmatrix} = 0$$

$\Rightarrow \vec{a}, \vec{b}, \vec{c}$ are coplanar vector.

$\Rightarrow \vec{p}, \vec{q}$ and $\vec{r}$ are also coplanar

$$\Rightarrow [\vec{p} \ \vec{q} \ \vec{r}] = 0$$

V01/1/006. A straight line passes through a point P whose position vector is $\vec{p}$ is parallel to the vector which is equally inclined to two unit vectors $\vec{a}$ and $\vec{b}$, in the plane of $\vec{a}$ and $\vec{b}$, then distance of point Q with position vector $\vec{q}$ from the line is :

(A) $|(\vec{q} - \vec{p}) \times (\vec{a} + \vec{b})|$

(B) $|(\vec{q} - \vec{p}) \cdot (\vec{a} + \vec{b})|$

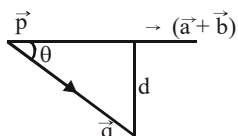
(C) $\frac{|(\vec{q} - \vec{p}) \times (\vec{a} + \vec{b})|}{|\vec{a} + \vec{b}|}$

(D) $\frac{|(\vec{q} - \vec{p}) \cdot (\vec{a} + \vec{b})|}{|\vec{a} + \vec{b}|}$

Ans.

C

Sol.

Equation of straight line . 

$$\vec{r} = \vec{p} + \lambda (\vec{a} + \vec{b})$$

$$d = |\vec{q} - \vec{p}| \sin \theta$$

$$(\vec{q} - \vec{p}) \times (\vec{a} + \vec{b}) = |\vec{q} - \vec{p}| |\vec{a} + \vec{b}| \sin \theta$$

$$d = \frac{|(\vec{q} - \vec{p}) \times (\vec{a} + \vec{b})|}{|\vec{a} + \vec{b}|}$$

V01/1/007. The non zero value of k for which the lines $kx - 4y + 7z + 16 = 0 = 4x + 3y - 2z + 3$ and $x - 3y + 4z + 6 = 0 = x - y + z + 1$ are intersecting is :

(A) 4

(B) 5

(C) 6

(D) 7

Ans. D**Sol.** If lines are intersecting all the 4 planes will have a common point.

$$\text{solving } 4x + 3y - 2z + 3 = 0$$

$$x - 3y + 4z + 6 = 0$$

$$x - y + z + 1 = 0$$

$$\text{we get } x = \frac{-1}{3}, y = -3, z = -\frac{11}{3}$$

$$\text{Put in } kx - 4y + 7z + 16 = 0$$

$$\text{we get } k = 7.$$

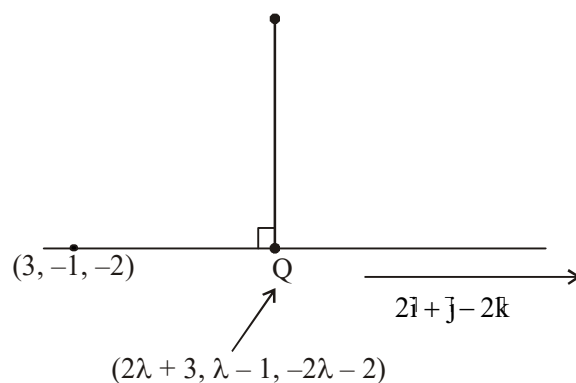
V01/1/008. A rigid body is spinning about the line $\vec{r} = 3\hat{i} - \hat{j} - 2\hat{k} + \lambda(2\hat{i} + \hat{j} - 2\hat{k})$ (which is the axis of rotation) with angular velocity of 5 radians per second, then the velocity of the particle at (4,1,0) is :

$$(A) \ 5(2\hat{i} - 2\hat{j} + \hat{k})$$

$$(B) \ 2\hat{i} - 2\hat{j} + \hat{k}$$

$$(C) \ \frac{2\hat{i} - \hat{j} + \hat{k}}{3}$$

$$(D) \ 4\hat{i} - 2\hat{j} + 2\hat{k}$$

Ans. A**Sol.**

$$\overrightarrow{PQ} \cdot (2\hat{i} + \hat{j} - 2\hat{k}) = 0$$

$$2(2\lambda - 1) + (\lambda - 2) + 2(2\lambda + 2) = 0$$

$$9\lambda = 0$$

$$\lambda = 0$$

$$Q(3, -1, -2)$$

$$\vec{r} = \overrightarrow{PQ} = \hat{i} + 2\hat{j} + 2\hat{k}$$

$$\Rightarrow |\overrightarrow{PQ}| = 3$$

$$\vec{v} = \vec{w} \times \vec{r}$$

$$|\vec{r}| = 3$$

$$|\vec{v}| = wr$$

$$\vec{w} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 1 & 2 & 2 \end{vmatrix}$$

$$= 5 \times 3$$

$$= 6\hat{i} - 6\hat{j} + 3\hat{k}$$

$$= 15$$

$$\Rightarrow \vec{v} = 15(\vec{w} \times \vec{r})$$

$$= 15 \cdot \frac{6\hat{i} - 6\hat{j} + 3\hat{k}}{9}$$

$$= 5(2\hat{i} - 2\hat{j} + \hat{k})$$

V01/1/009. Let $2\hat{a} + \vec{b} = \hat{a} \times \vec{b}$ (where $\hat{a}$ is a unit vector) then $|3\vec{b} + 4\hat{a}|$ is equal to :

(A) 0

(B) 1

(C) 2

(D) 3

Ans. C

Sol. $2\hat{a} + \vec{b} = \hat{a} \times \vec{b}$ _____(i)

Taking dot with $\hat{a} \times \hat{b}$

$$\Rightarrow (\hat{a} \times \hat{b})(\hat{a} \times \hat{b}) = 0 \Rightarrow \vec{b} = \lambda \hat{a}$$

from (i)

$$\Rightarrow \lambda = -2 \Rightarrow \vec{b} = -2\hat{a}$$

$$\Rightarrow |3\vec{b} + 4\hat{a}| = |-6\hat{a} + 4\hat{a}|$$

$$= |-2\hat{a}| = 2$$

V01/1/010. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three non zero vectors such that projection of $(\vec{b} + \vec{c})$ on $\vec{a}$ is equal to $2|\vec{a}|$, projection

of $(\vec{a} + \vec{c})$ on $\vec{b}$ is $3|\vec{b}|$ and projection of $(\vec{a} + \vec{b})$ on $\vec{c}$ is $4|\vec{c}|$, (where $|\vec{a}|, |\vec{b}|, |\vec{c}| \in \mathbb{N}$), then the minimum value of $|\vec{a} + \vec{b} + \vec{c}|$ is equal to :

- (A) 12 (B) $2\sqrt{3}$ (C) $3\sqrt{2}$ (D) 18

Ans. B

Sol.
$$\frac{(\vec{b} + \vec{c}) \cdot \vec{a}}{|\vec{a}|} = 2|\vec{a}|$$

$$\Rightarrow (\vec{b} + \vec{c}) \cdot \vec{a} = 2a^2 \quad \dots\dots\dots(1)$$

$$(\vec{c} + \vec{a}) \cdot \vec{b} = 3b^2 \quad \dots\dots\dots(2)$$

$$(\vec{a} + \vec{b}) \cdot \vec{c} = 4c^2 \quad \dots\dots\dots(3)$$

add the equations

$$2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 2a^2 + 3b^2 + 4c^2$$

$$|\vec{a} + \vec{b} + \vec{c}|^2 = \Sigma a^2 + 2\Sigma \vec{a} \cdot \vec{b} = 3a^2 + 4b^2 + 5c^2$$

$$|\vec{a} + \vec{b} + \vec{c}| = \sqrt{3a^2 + 4b^2 + 5c^2}$$

$$\text{minimum} = \sqrt{12} = 2\sqrt{3}$$

V01/1/011. If the intercept made by the plane $\vec{r} \cdot \vec{n} = q$ on x, y, z axis are respectively are a_1, a_2 & a_3 respectively, then :

(A) $\vec{n} = (qa_1)\hat{i} + (qa_2)\hat{j} + (qa_3)\hat{k}$

(B) $\vec{n} = \frac{a_1}{q}\hat{i} + \frac{a_2}{q}\hat{j} + \frac{a_3}{q}\hat{k}$

(C) $\vec{n} = \frac{q}{a_1}\hat{i} + \frac{q}{a_2}\hat{j} + \frac{q}{a_3}\hat{k}$

(D) $\vec{n} = \left(\frac{1}{a_1 q}\right)\hat{i} + \left(\frac{1}{a_2 q}\right)\hat{j} + \left(\frac{1}{a_3 q}\right)\hat{k}$

Ans. C

Sol. for x axis $\vec{r} = a_1\hat{i} \Rightarrow \vec{n} \cdot \hat{i} = \frac{q}{a_1}$

$$\vec{n} \cdot \hat{j} = \frac{q}{a_2}$$

$$\vec{n} \cdot \hat{k} = \frac{q}{a_3}$$

$$\vec{n} = (\vec{n} \cdot \hat{i})\hat{i} + (\vec{n} \cdot \hat{j})\hat{j} + (\vec{n} \cdot \hat{k})\hat{k}$$

$$\vec{n} = \frac{q}{a_1}\hat{i} + \frac{q}{a_2}\hat{j} + \frac{q}{a_3}\hat{k}$$

V01/1/012. OABC is a tetrahedron. The position vectors of A, B and C are $\hat{i}, \hat{i} + \hat{j}, \hat{j} + \hat{k}$ respectively. 'O' is the origin. The distance of the plane face ABC from origin O is :

- (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{2\sqrt{2}}$ (D) 1

Ans. B

Sol. Volume of tetrahedron

$$= \frac{1}{6} [\vec{OA} \vec{OB} \vec{OC}] = \frac{1}{6} \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = \frac{1}{6}$$

$$\text{Area of face ABC} = \frac{1}{2} |(\hat{i} + \hat{j} - \hat{i}) \times (\hat{j} + \hat{k} - \hat{i})|$$

$$= \frac{1}{2} |\hat{i} + \hat{k}| = \frac{1}{\sqrt{2}}$$

$\therefore$ required distance

$$= \frac{3 \times \text{volume}}{\text{Area of base}} = \frac{3\sqrt{2}}{6} = \frac{1}{\sqrt{2}}$$

V01/1/013. If $\vec{p}$ and $\vec{q}$ are two diagonals of a quadrilateral such that $|\vec{p} - \vec{q}| = \vec{p} \cdot \vec{q}, |\vec{p}| = 1, |\vec{q}| = \sqrt{2}$, then area of quadrilateral is equal to :

- (A) 1 (B) $\frac{1}{2}$ (C) $\frac{3}{2}$ (D) $\sqrt{3}$

Ans. B

Sol. $|\vec{p} - \vec{q}| = \vec{p} \cdot \vec{q}$

$$|\vec{p}|^2 + |\vec{q}|^2 - 2\vec{p} \cdot \vec{q} = 2 \cos^2 \theta$$

$$\cos \theta = \frac{1}{\sqrt{2}}$$

$$\text{Area} = \frac{1}{2} |\vec{p}| |\vec{q}| \sin \theta = \frac{1}{2}$$

V01/1/014. If $\vec{a} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + 4\hat{j} + 3\hat{k}$ are one of the sides and medians respectively, of a triangle through the same vertex, then area of triangle is :

(A) $\sqrt{86}$

(B) $\frac{1}{2}\sqrt{85}$

(C) $\frac{1}{2}\sqrt{84}$

(D) $\sqrt{80}$

Ans. A

Sol. $\vec{a} = \hat{i} - \hat{j} + \hat{k}$

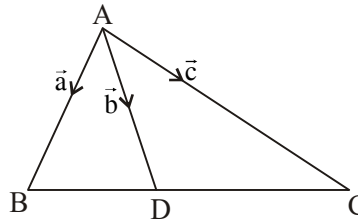
$\vec{b} = 2\hat{i} + 4\hat{j} + 3\hat{k}$

$\vec{b} = \frac{\vec{a} + \vec{c}}{2}$

$\therefore \text{Area} = \frac{1}{2} |\vec{a} \times \vec{c}|$

$= \frac{1}{2} |\vec{a} \times (2\vec{b} - \vec{a})|$

$= |\vec{a} \times \vec{b}| = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 1 \\ 2 & 4 & 3 \end{vmatrix} = \sqrt{86}$



V01/1/015. Let OABCDE is a regular hexagon, O is origin and position vector of A and B are $\vec{a}$ and $\vec{b}$ respectively. The position vector of point of intersection of OB and AC is :

(A) $\frac{2}{3}\vec{b}$

(B) $\frac{2}{3}(\vec{a} + \vec{b})$

(C) $\frac{\vec{b}}{3}$

(D) $\vec{a} - \frac{\vec{b}}{3}$

Ans. A

Sol. $\vec{AB} = \vec{b} - \vec{a}$

observe that in regular hexagon

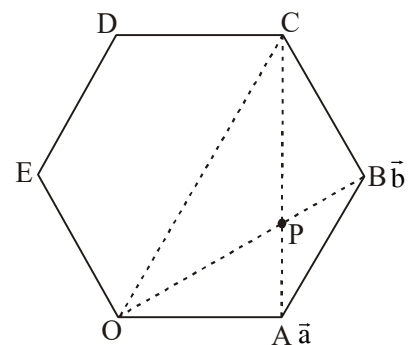
length of OC is double the length of its side and $OC \parallel AB$

$\Rightarrow \vec{OC} = 2(\vec{b} - \vec{a})$

Line OB : $\vec{r} = \lambda \vec{b}$

Line AC : $\vec{r} = \vec{a} + \mu(2\vec{b} - 2\vec{a} - \vec{a})$

For P $\lambda = 2\mu, 1 - 3\mu = 0$



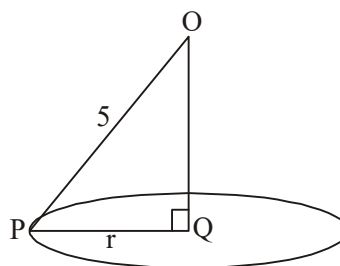
$$\Rightarrow \vec{r} = \frac{2}{3}\vec{b}$$

V01/1/016. Let P is a variable point moves such that $OP = 5$ and $AP = 13$. If $A(6, 12, 4)$, O is origin and length of path traced by P is $\frac{2\pi}{7}\sqrt{L}$, then :

- (A) $L = 1024$ (B) $L = 1056$ (C) $L = 1065$ (D) $L = 1000$

Ans. B

Sol. Let moving point be $P(x, y, z)$
 $\Rightarrow (x - 6)^2 + (y - 12)^2 + (z - 4)^2 = 13^2$ (sphere)
 and $x^2 + y^2 + z^2 = 25$ (sphere)
 $\Rightarrow 12x + 24y + 8z - 196 = 25 - 169$
 $\Rightarrow 12x + 24y + 8z = 52$
 $3x + 6y + 2z = 13$



Clearly locus of P is common plane of two spheres
 i.e. circles shown in diagram

$$OQ = \frac{13}{7} \quad r = \sqrt{25 - \left(\frac{13}{7}\right)^2} = \frac{1}{7}\sqrt{1056}$$

$$L = \frac{2\pi}{7}\sqrt{1056}$$

V01/1/017. If the four plane faces of a tetrahedron are represented by the equation $\vec{r} \cdot (\hat{\ell}\hat{i} + m\hat{j}) = 0$, $\vec{r} \cdot (m\hat{j} + n\hat{k}) = 0$, $\vec{r} \cdot (n\hat{k} + \ell\hat{i}) = 0$, $\vec{r} \cdot (\ell\hat{i} + m\hat{j} + n\hat{k}) = p$, then the volume of the tetrahedron, is :

- (A) $\frac{p^3}{6lmn}$ (B) $\frac{2p^3}{3lmn}$ (C) $\frac{3p^3}{lmn}$ (D) $\frac{6p^3}{lmn}$

Ans. B

Sol. The first three planes meet at the point whose position vector is $(0, 0, 0)$. The first two and the fourth planes meet at the point whose position vector is $\left(\frac{p}{1}, \frac{-p}{m}, \frac{p}{n}\right)$. Similarly, the other two vertices of the tetrahedron have position vectors $\left(\frac{-p}{1}, \frac{p}{m}, \frac{p}{n}\right)$ and $\left(\frac{p}{1}, \frac{p}{m}, \frac{-p}{n}\right)$. Hence, the volume of the tetrahedron is

$$\frac{1}{6} \begin{vmatrix} p/1 & -p/m & p/n \\ -p/1 & p/m & p/n \\ p/1 & p/m & -p/n \end{vmatrix} = \frac{p^3}{6lmn} \begin{vmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= \frac{p^3}{6lmn} \begin{vmatrix} 0 & 0 & 2 \\ 0 & 2 & 0 \\ 1 & 1 & -1 \end{vmatrix} = \frac{4p^3}{6lmn} = \frac{2p^3}{3lmn}.$$

- V01/1/018. A line is inclined at angles α, β, γ with the positive direction of x, y, z axes respectively. If $\beta + \gamma = \frac{\pi}{2}$, then the maximum possible value of $(\cos\alpha + \sin\beta + \sin\gamma)^2$ is equal to :
- (A) 1 (B) 2 (C) 3 (D) None of these

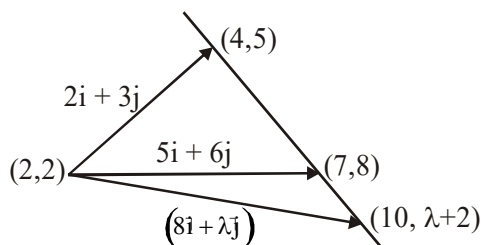
Ans. B

Sol. $\beta + \gamma = \frac{\pi}{2}$
 $\cos\beta = \sin\gamma$
 $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$
 $\cos^2\alpha = 0$
 $(\cos\alpha + \sin\beta + \sin\gamma)^2$
 $= \sin^2\beta + \sin^2\gamma + 2\sin\beta\sin\gamma$
 $= 1 + \sin 2\beta)_{\max} = 2$

- V01/1/019. The vectors $2\hat{i} + 3\hat{j}$, $5\hat{i} + 6\hat{j}$ and $8\hat{i} + \lambda\hat{j}$ have their initial point at (2, 2). The value of λ so that the vectors terminate on one straight line is :
- (A) 0 (B) 3 (C) 8 (D) 9

Ans. D

Sol.

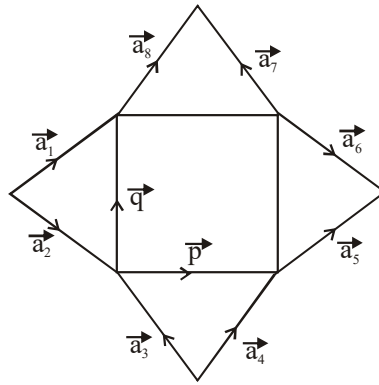


Clearly, points (4, 5), (7, 8) and (10, $\lambda + 2$) are collinear

$$\begin{vmatrix} 4 & 5 & 1 \\ 7 & 8 & 1 \\ 10 & \lambda + 2 & 1 \end{vmatrix} = 0$$

$$\lambda = 9$$

V01/1/020. Figure shows a square. On each side of a square an outward equilateral triangle is constructed, $\vec{a}_1, \vec{a}_2, \vec{a}_3, \vec{a}_4, \vec{a}_5, \vec{a}_6, \vec{a}_7, \vec{a}_8, \vec{p}$ & $\vec{q}$ are as shown in figure. If $\vec{a}_2 \times \vec{a}_1 + \vec{a}_4 \times \vec{a}_3 + \vec{a}_6 \times \vec{a}_5 + \vec{a}_8 \times \vec{a}_7 = \lambda \vec{p} \times \vec{q}$, then λ is equal to :



- (A) $\sqrt{3}$ (B) $\frac{\sqrt{3}}{2}$ (C) $2\sqrt{3}$ (D) $\frac{\sqrt{3}}{4}$

Ans. C

Sol. $\vec{a}_{i+1} \times \vec{a}_i = \frac{\sqrt{3}}{2} l^2 \hat{n}$

$$\vec{p} \times \vec{q} = l^2 \hat{n}$$

$$2\sqrt{3}l^2 \hat{n} = \lambda l^2 \hat{n}$$

$$\lambda = 2\sqrt{3}$$

($\hat{n}$ is unit vector $\perp$ to plane and ' l ' is length of square)

V01/1/021. In cartesian plane the position vectors of vertices of ΔABC are $\vec{a}, \vec{b}, \vec{c}$ such that $\vec{a} \cdot \vec{a} = \vec{b} \cdot \vec{b} = \vec{c} \cdot \vec{c}$. If $\vec{a} = \lambda \vec{b} + \mu \vec{c}$ (where λ & μ are real), then position vector of orthocentre of ΔABC is :

- (A) $\vec{a} + \vec{b} + \vec{c}$ (B) $\vec{b} + \vec{c} - \vec{a}$

(C) $\frac{\vec{a} + \vec{b} + \vec{c}}{3}$

(D) $\frac{\vec{a} + \vec{b} + \vec{c}}{2}$

Ans. A

Sol. Circumcentre is origin and centroid $\frac{\vec{a} + \vec{b} + \vec{c}}{3}$

$\therefore$ centroid divides orthocentre and circumcentre in 2 : 1

$\Rightarrow$ orthocentre is $\vec{a} + \vec{b} + \vec{c}$

V01/1/022. Consider the lines

$L_1: x = 3 - t, y = 2 + t, z = 5t$, intersecting the plane $x - y + 2z = 9$ at the point A

$L_2: x = 1 + 2t, y = 4t, z = 2 - 3t$, intersecting the plane $x + 2y - z + 1 = 0$ at the point B

and $L_3: x = y - 1 = 2z$, intersecting the plane $4x - y + 3z = 8$ at the point C. The points A, B, C :

(A) constitute a right triangle

(B) constitute an acute triangle

(C) constitute an obtuse triangle.

(D) do not form a triangle.

Ans. B

Sol. $L_1: x = 3 - t, y = 2 + t, z = 5t$

Plane : $x - y + 2z = 9$

$$(3 - t) - (2 + t) + 10t = 9$$

$$3 - t - 2 - t + 10t = 9$$

$$8t = 8 \Rightarrow t = 1$$

$$A \equiv (2, 3, 5)$$

$L_2: x = 1 + 2t, y = 4t, z = 2 - 3t$

Plane : $x + 2y - z + 1 = 0$

$$1 + 2t + 8t - 2 + 3t + 1 = 0$$

$$t = 0$$

$$B \equiv (1, 0, 2)$$

$$L_3: \frac{x}{2} = \frac{y-1}{2} = z = \lambda$$

$$x = 2\lambda, y = 2\lambda + 1, z = \lambda$$

Plane : $4x - y + 3z = 8$

$$8\lambda - 2\lambda - 1 + 3\lambda = 8$$

$$9\lambda = 9$$

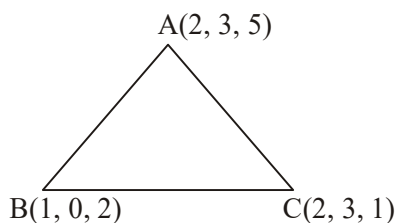
$$\lambda = 1$$

$$C \equiv (2, 3, 1)$$

$$AB: \sqrt{19}, \quad BC: \sqrt{11}, \quad CA = 4$$

and $AB + BC > CA$

$\therefore$ A, B, C constitute an acute triangle



V01/1/023. Given the position vectors of the vertices of a triangle ABC, $A \equiv (\bar{a})$; $B \equiv (\bar{b})$; $C \equiv (\bar{c})$. A vector $\vec{r}$ is parallel to the altitude drawn from the vertex A, making an obtuse angle with the positive Y-axis.

If $|\vec{r}| = 2\sqrt{34}$; $\bar{a} = 2\hat{i} - \hat{j} - 3\hat{k}$; $\bar{b} = \hat{i} + 2\hat{j} - 4\hat{k}$; $\bar{c} = 3\hat{i} - \hat{j} - 2\hat{k}$, then $\vec{r}$ is :

- (A) $-6\hat{i} - 8\hat{j} - 6\hat{k}$ (B) $6\hat{i} - 8\hat{j} + 6\hat{k}$ (C) $-6\hat{i} - 8\hat{j} + 6\hat{k}$ (D) $6\hat{i} + 8\hat{j} + 6\hat{k}$

Ans. A

Sol. let pt p be

$$(1 + 2\lambda, 2 - 3\lambda, -4 + 2\lambda)$$

$$\overrightarrow{AP} \cdot \overrightarrow{BC} = 0$$

$$((2\lambda - 1)\hat{i} + (3 - 3\lambda)\hat{j} + (2\lambda - 1)\hat{k}) \cdot (2\hat{i} - 3\hat{j} + 2\hat{k}) = 0$$

$$4\lambda - 2 - 9 + 9\lambda + 4\lambda - 2 = 0$$

$$17\lambda - 13 = 0$$

$$\lambda = \frac{13}{17}$$

$$\overrightarrow{AP} = 9\hat{i} + 12\hat{j} + 9\hat{k}$$

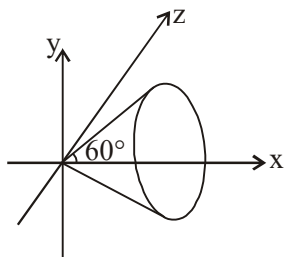
$$\vec{r} = -6\hat{i} - 8\hat{j} - 6\hat{k}$$

V01/1/024. Locus of the point P , for which $\vec{OP}$ represents a vector with direction cosine $\cos \alpha = \frac{1}{2}$ ('O' is the origin) is:

- (A) A circle parallel to y-z plane with centre on the x-axis
 (B) a cone concentric with positive x-axis having vertex at the origin and the slant height equal to the magnitude of the vector
 (C) a ray emanating from the origin and making an angle of 60° with x-axis
 (D) a disc parallel to y-z plane with centre on x-axis & radius equal to $\left| \vec{OP} \right| \sin 60^\circ$

Ans. B

Sol. From the diagram, its is obvious that locus is a cone concentric with the positive x-axis having vertex at the Origin and the slant height equal to the magnitude of vector.



V01/1/025. Given $|\vec{p}| = 2$; $|\vec{q}| = 3$ and $\vec{p} \cdot \vec{q} = 0$. If $\vec{V} = (\vec{p} \times (\vec{p} \times (\vec{p} \times (\vec{p} \times \vec{q}))))$ then the vector $\vec{V}$ is :

(A) collinear with $\vec{p}$

(B) $\vec{V} = 16\vec{p}$

(C) $\vec{V} = 48\vec{q}$

(D) $\vec{V} = 16\vec{q}$

Ans. D

Sol. Given $|\vec{p}| = 2$; $|\vec{q}| = 3$ and $\vec{p} \cdot \vec{q} = 0$

$$\text{If } \vec{v} = (\vec{p} \times (\vec{p} \times (\vec{p} \times (\vec{p} \times \vec{q}))))$$

$$= \vec{p} \times \vec{p} \times (\vec{p} \times 6\hat{n})$$

$$= \vec{p} \times \vec{p} \times 2 \times 6\hat{n}_1$$

$$= \vec{p} \times (\vec{p} \times 12\hat{n}_1)$$

$$= \vec{p} \times 24\hat{n}_2$$

$$= 48\hat{n}_3$$

$$= 48\hat{q}$$

$$= 48 \cdot \frac{\vec{q}}{|\vec{q}|} = \frac{48}{3} \hat{q} = 16\hat{q}$$

V01/1/026. If the points with position vectors $A(\vec{a}) = (2-x)\hat{i} + 2\hat{j} + 2\hat{k}$, $B(\vec{b}) = 2\hat{i} + (2-y)\hat{j} + 2\hat{k}$,

$C(\vec{c}) = 2\hat{i} + 2\hat{j} + (2-z)\hat{k}$ and $D(\vec{d}) = \hat{i} + \hat{j} + \hat{k}$ are coplanar, then :

(A) $\frac{1}{x-2} + \frac{1}{y-2} + \frac{1}{z-2} = 1$

(B) $\frac{1}{1-x} + \frac{1}{1-y} + \frac{1}{1-z} = 1$

(C) $x + y + z = 1$

(D) $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$

Ans. D

Sol. $\overrightarrow{DA}, \overrightarrow{DB}, \overrightarrow{DC}$ are coplanar

$$\Rightarrow \begin{vmatrix} 1-x & 1 & 1 \\ 1 & 1-y & 1 \\ 1 & 1 & 1-z \end{vmatrix} = 0$$

$$R_1 \rightarrow R_1 - R_3, R_2 \rightarrow R_2 - R_3$$

$$\begin{vmatrix} -x & 0 & z \\ 0 & -y & z \\ 1 & 1 & 1-z \end{vmatrix} = 0$$

$$-x[-y + yz - z] + z[0 + y] = 0$$

$$xy + xz + yz = xyz$$

$$\Rightarrow \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$$

V01/1/027. Four vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{x}$ satisfy the relation $(\vec{a} \cdot \vec{x})\vec{b} = \vec{c} + \vec{x}$ where $\vec{b} \cdot \vec{a} \neq 1$. The value of $\vec{x}$ in terms of $\vec{a}$, $\vec{b}$ and $\vec{c}$ is equal to :

(A) $\frac{(\vec{a} \cdot \vec{c})\vec{b} - \vec{c}(\vec{a} \cdot \vec{b} - 1)}{(\vec{a} \cdot \vec{b} - 1)}$

(B) $\frac{\vec{c}}{\vec{a} \cdot \vec{b} - 1}$

(C) $\frac{2(\vec{a} \cdot \vec{c})\vec{b} + \vec{c}}{\vec{a} \cdot \vec{b} - 1}$

(D) $\frac{2(\vec{a} \cdot \vec{c})\vec{c} + \vec{c}}{\vec{a} \cdot \vec{b} - 1}$

Ans. A

Sol. $(\vec{a} \cdot \vec{x})\vec{b} = \vec{c} + \vec{x}$ _____(1)

taking dot product with $\vec{a}$

$$(\vec{a} \cdot \vec{x})(\vec{b} \cdot \vec{a}) = (\vec{c} \cdot \vec{a}) + (\vec{a} \cdot \vec{x})$$

$$\Rightarrow (\vec{a} \cdot \vec{x})(\vec{b} \cdot \vec{a} - 1) = \vec{c} \cdot \vec{a}$$

$$\Rightarrow \vec{a} \cdot \vec{x} = \frac{\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a} - 1}$$
 _____(2)

from (1) and (2)

$$\begin{aligned} \left(\frac{\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a} - 1} \right) \vec{b} &= \vec{c} + \vec{x} \\ \Rightarrow \vec{x} &= \left(\frac{\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a} - 1} \right) \vec{b} - \vec{c} \\ &= \frac{(\vec{a} \cdot \vec{c}) \vec{b} - \vec{c} (\vec{a} \cdot \vec{b} - 1)}{(\vec{a} \cdot \vec{b} - 1)} \end{aligned}$$

V01/1/028. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are three unit vectors and α, β, γ are the angles between the vectors $\vec{a}$, $\vec{b}$; $\vec{b}$, $\vec{c}$ and $\vec{c}$, $\vec{a}$ respectively and also α, β, γ are the angles which the unit vector $\hat{i} + m\hat{j} + n\hat{k}$ makes with the positive x-axis, y-axis and z-axis respectively, then the value of $|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|$ equals :

- (A) $|\sum \ell - 1|$ (B) $2\sum(lm - 1)$
 (C) $-2\sum lm + 2\sum l + 2$ (D) $2(1 - \sum lm)$

Ans.

A

Sol.

$$(\ell + m + n)^2 = 1 + 2(\ell n + mn + nl)$$

$$\vec{a} \cdot \vec{b} = \cos \alpha = \ell$$

$$\vec{b} \cdot \vec{c} = \cos \beta = m$$

$$\vec{c} \cdot \vec{a} = \cos \gamma = n$$

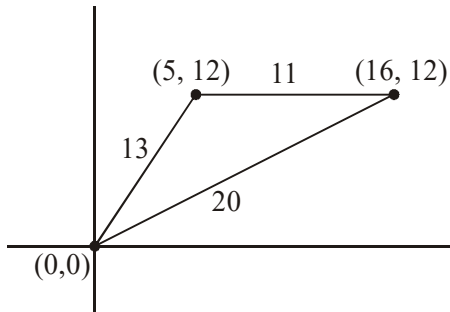
$$\begin{aligned} |\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}|^2 &= |\vec{a} \times \vec{b}|^2 + |\vec{b} \times \vec{c}|^2 + |\vec{c} \times \vec{a}|^2 + 2(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{c}) + 2(\vec{b} \times \vec{c}) \cdot (\vec{c} \times \vec{a}) + 2(\vec{c} \times \vec{a}) \cdot (\vec{a} \times \vec{b}) \\ &= \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma + 2[\cos \alpha \cos \beta + \cos \beta \cos \gamma + \cos \gamma \cos \alpha - \cos \alpha - \cos \beta - \cos \gamma] \\ &= 1 - \cos^2 \alpha + 1 - \cos^2 \beta + 1 - \cos^2 \gamma + 2[\ell m + mn + n\ell - \ell - m - n] \\ &= 3 - (\ell^2 + m^2 + n^2) + 2(\ell m + mn + n\ell) - 2(\ell + m + n) \\ &= 2 - 2(\ell m + mn + n\ell) - 2(\ell + m + n) \\ &= 1 + 2(\ell m + mn + n\ell) - (\ell + m + n)^2 \\ &= 1 + (l + m + n)^2 - 2(l + m + n) \\ &= (1 - \sum \ell)^2 \end{aligned}$$

V01/1/029. The incentre of the triangle formed by $(0, 0)$, $(5, 12)$, $(16, 12)$ is $(\alpha, 9)$. If $\vec{a} + 5\vec{b} = \vec{c}$ and $\vec{a} - \alpha\vec{b} = 2\vec{c}$, then:

- (A) $\vec{a}$ and $\vec{c}$ are like but $\vec{b}$ and $\vec{c}$ are unlike vectors
 (B) $\vec{a}$ and $\vec{b}$ are unlike vectors and so also $\vec{a}$ and $\vec{c}$
 (C) $\vec{b}$ and $\vec{c}$ are like but $\vec{a}$ and $\vec{b}$ are unlike vectors
 (D) $\vec{a}$ and $\vec{c}$ are unlike vectors and so also $\vec{b}$ and $\vec{c}$

Ans. A

Sol.



$$I\left(\frac{100 + 208 + 0}{20 + 13 + 11}, 9\right)$$

$$I(7, 9)$$

$$\alpha = 7$$

$$\vec{a} + 5\vec{b} = \vec{c}$$

$$\vec{a} - 7\vec{b} = 2\vec{c}$$

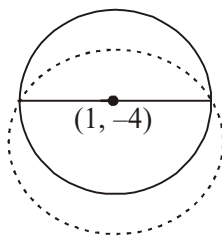
$$\Rightarrow 12\vec{b} = -\vec{c} \text{ \& } 12\vec{a} = 17\vec{c}$$

V01/1/030. If the circle $x^2 + y^2 + 4x + 22y + c = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - d = 0$. If the lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{4-z}{\lambda}$ and $\frac{x-1}{\lambda} = \frac{y-4}{2} = \frac{z-\alpha}{1}$ intersect, then the value of $\alpha\lambda$ is (Where, $\alpha\lambda$ is equal to $\frac{c+d}{10}$):

- (A) 1 (B) 2 (C) $\frac{1}{2}$ (D) 0

Ans. D

Sol.



Common chord

$$6x + 14y + c + d = 0$$

$$(1, -4)$$

$$6 - 56 + c + d = 0$$

$$\Rightarrow c + d = 50$$

$$\lambda = \frac{c+d}{10}$$

$$\lambda = 5$$

Lines are intersecting

$$\Rightarrow (\vec{b} - \vec{a}) \cdot (\vec{P}_1 \times \vec{P}_2) = 0$$

$$\begin{vmatrix} 1 & -1 & -1 \\ 1 & 1 & -\lambda \\ \lambda & 2 & 1 \end{vmatrix} = 0$$

$$(1 + 2\lambda) + (1 + \lambda^2) - (2 - \lambda) = 0$$

$$\lambda^2 + 3\lambda = 0$$

$$\Rightarrow \lambda = 0, -3$$

V01/1/031. The minimum value of $x^2 + y^2 + z^2$ if $ax + by + cz = p$, is :

- (A) $\left(\frac{p}{a+b+c}\right)^2$ (B) $\frac{p^2}{a^2+b^2+c^2}$
(C) $\frac{a^2+b^2+c^2}{p^2}$ (D) 0

Ans. B

Sol. Perpendicular distance from origin

$$d = \frac{p}{\sqrt{a^2+b^2+c^2}} \quad \therefore d^2 = \frac{p^2}{a^2+b^2+c^2}$$

V01/1/032. If the angle between the vectors $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{3}$ and the area of the triangle with adjacent sides parallel to $\vec{a}$ and $\vec{b}$ is 3, then $\vec{a} \cdot \vec{b} =$:

- (A) $\sqrt{3}$ (B) $2\sqrt{3}$ (C) $4\sqrt{3}$ (D) $\frac{\sqrt{3}}{2}$

Ans. B

Sol. Area of triangle = $\frac{1}{2} |\vec{a} \times \vec{b}| = 3$

$$\Rightarrow |\vec{a}| |\vec{b}| \sin \frac{\pi}{3} = 6 \quad \Rightarrow |\vec{a}| |\vec{b}| = \frac{12}{\sqrt{3}}$$

$$\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \frac{\pi}{3} = 2\sqrt{3}$$

V01/1/033. If acute angle between the line $\vec{r} = \hat{i} + 2\hat{j} + \lambda(4\hat{i} - 3\hat{k})$ and xy plane is θ_1 and acute angle between the planes $x + 2y = 0$ and $2x + y = 0$ is θ_2 then $(\cos^2\theta_1 + \sin^2\theta_2)$ equals :

- (A) 1 (B) $\frac{1}{4}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

Ans. A

Sol. $\cos\theta_1 = \frac{4}{5}$

$$\cos\theta_2 = \frac{4}{5}$$

$$\Rightarrow \cos^2\theta_1 + \sin^2\theta_2 = 1$$

V01/1/034. If $\vec{a}$ and $\vec{b}$ are non-zero, non-collinear vectors such that $|\vec{a}| = 2$, $\vec{a} \cdot \vec{b} = 1$ and angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{3}$.

If $\vec{r}$ is any vector such that $\vec{r} \cdot \vec{a} = 2$, $\vec{r} \cdot \vec{b} = 8$, $(\vec{r} + 2\vec{a} - 10\vec{b}) \cdot (\vec{a} \times \vec{b}) = 4\sqrt{3}$ and $\vec{r} + 2\vec{a} - 10\vec{b} = \lambda (\vec{a} \times \vec{b})$ then λ is equal to :

- (A) $\frac{1}{2}$ (B) 2 (C) $\frac{1}{4}$ (D) None of these

Ans. D

Sol. $\lambda (\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{b}) = 4\sqrt{3} \quad \left(ab \cos \frac{\pi}{3} = 1 \right)$

$$\lambda (a^2 b^2 - (\vec{a} \cdot \vec{b})^2) = 4\sqrt{3} \quad \Rightarrow b = 1$$

$$\lambda (4 \times 1 - (1)^2) = 4\sqrt{3}$$

$$\lambda = \frac{4\sqrt{3}}{3}$$

- V01/1/035. Let $\vec{a} = 3\hat{i} + 2\hat{j} + 4\hat{k}$; $\vec{b} = 2(\hat{i} + \hat{k})$ and $\vec{c} = 4\hat{i} + 2\hat{j} + 3\hat{k}$. Sum of the values of 'α' for which the equation $x\vec{a} + y\vec{b} + z\vec{c} = \alpha(x\hat{i} + y\hat{j} + z\hat{k})$ has non trivial solution is :
- (A) -1 (B) 4 (C) 7 (D) 8

Ans. C

Sol. $x(3\hat{i} + 2\hat{j} + 4\hat{k}) + y(2\hat{i} + 2\hat{k}) + z(4\hat{i} + 2\hat{j} + 3\hat{k}) = \alpha(x\hat{i} + y\hat{j} + z\hat{k})$
 $\Rightarrow (3 - \alpha)x + 2y + 4z = 0$
 $2x - \alpha y + 2z = 0$
 $4x + 2y + (3 - \alpha)z = 0$
 For non-trivial solution

$$\begin{vmatrix} 3-\alpha & 2 & 4 \\ 2 & -\alpha & 2 \\ 4 & 2 & 3-\alpha \end{vmatrix} = 0$$

$\Rightarrow \alpha^3 - 6\alpha^2 - 15\alpha - 8 = 0$
 $\Rightarrow (\alpha + 1)^2 (\alpha - 8) = 0$
 $\Rightarrow \alpha = 8, -1$

- V01/1/036. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 1, |\vec{b}| = 4, \vec{a} \cdot \vec{b} = 2$. If $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$ then angle between $\vec{b}$ and $\vec{c}$ is :
- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{2\pi}{3}$ (D) $\frac{5\pi}{6}$

Ans. D

Sol. $|\vec{c}|^2 = 4(\vec{a} \times \vec{b})^2 + 9b^2 = 4(a^2b^2 - (\vec{a} \cdot \vec{b})^2) + 9b^2 = 192$

$\vec{c} + 3\vec{b} = 2\vec{a} \times \vec{b} \quad \Rightarrow c^2 + 9b^2 + 6\vec{b} \cdot \vec{c} = 4(a^2b^2 - (\vec{a} \cdot \vec{b})^2)$

$\Rightarrow 6 \cdot 4 \cdot \sqrt{192} \cos \theta = -288 \Rightarrow \cos \theta = \frac{-\sqrt{3}}{2}$

- V01/1/037. If $\vec{a}, \vec{b}, \vec{c}$ are unit vectors, then the value of $|\vec{a} - 2\vec{b}|^2 + |\vec{b} - 2\vec{c}|^2 + |\vec{c} - 2\vec{a}|^2$ does not exceed :
- (A) 9 (B) 12 (C) 18 (D) 21

Ans. D

Sol. $|\vec{a} - 2\vec{b}|^2 + |\vec{b} - 2\vec{c}|^2 + |\vec{c} - 2\vec{a}|^2$

$$\begin{aligned}
&= 5a^2 + 5b^2 + 5c^2 - 4(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \\
&= 15 - 4\left(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}\right) \leq 15 - 4\left(\frac{-3}{2}\right) = 21 \\
\therefore \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} &\geq \frac{-3}{2}
\end{aligned}$$

V01/1/038. The adjacent side vectors $\overrightarrow{OA}$ and $\overrightarrow{OB}$ of a rectangle OACB are $\vec{a}$ and $\vec{b}$ respectively where O is the origin. If $16|\vec{a} \times \vec{b}| = 3(|\vec{a}| + |\vec{b}|)^2$ and θ be the acute angle between the diagonals OC and AB then the possible value of $\tan(\theta/2)$ is :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{3}$

Ans. D

Sol. $16|\vec{a}||\vec{b}|\sin\frac{\pi}{2} = 3|\vec{a}|^2 + 3|\vec{b}|^2 + 6|\vec{a}||\vec{b}|$
 $\Rightarrow 3a^2 - 10ab + 3b^2 = 0 \quad \Rightarrow (3a - b)(a - 3b) = 0$

Now $\vec{OC} \cdot \vec{AB} = (\vec{a} + \vec{b}) \cdot (\vec{b} - \vec{a}) = |\vec{OC}||\vec{AB}|\cos\theta$

$$\Rightarrow \frac{b^2 - a^2}{\sqrt{a^2 + b^2}\sqrt{a^2 + b^2}} = \cos\theta = \frac{9a^2 - a^2}{9a^2 + a^2} \text{ \{using } b = 3a\}$$

$$\therefore \cos\theta = \frac{4}{5} \quad \therefore \tan\theta/2 = \sqrt{\frac{1 - \cos\theta}{1 + \cos\theta}} = \sqrt{\frac{1 - 4/5}{1 + 4/5}} = \frac{1}{3}$$

V01/1/039. Let $\vec{a}_r = x_r\hat{i} + y_r\hat{j} + z_r\hat{k}$, $r = 1, 2, 3$ be three mutually perpendicular unit vectors, then the value of

$$\begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix} \text{ is equal to :}$$

(A) 0 (B) ± 1 (C) ± 2 (D) ± 4

Ans. B

Sol. $\vec{a}_1 = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$, $\vec{a}_2 = x_2\hat{i} + y_2\hat{j} + z_2\hat{k}$ and $\vec{a}_3 = x_3\hat{i} + y_3\hat{j} + z_3\hat{k}$ are mutually perpendicular unit

verctors, then $[\vec{a}_1 \vec{a}_2 \vec{a}_3] = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = \pm 1$

V01/1/040. Consider matrices $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 1 & 2 \\ 1 & -1 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 2 & 1 & 3 \\ 4 & 1 & -1 \\ 2 & 2 & 3 \end{bmatrix}$; $C = \begin{bmatrix} 14 \\ 12 \\ 2 \end{bmatrix}$; $D = \begin{bmatrix} 13 \\ 11 \\ 14 \end{bmatrix}$; $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ such that solutions

of equation $AX = C$ and $BX = D$ represents two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ respectively in three dimensional space. If $P'Q'$ is the reflection of the line PQ in the plane $\Pi : x + y + z = 9$, then the point which does not lie on $P'Q'$ is :

- (A) (3, 4, 2) (B) (5, 3, 4) (C) (7, 2, 3) (D) (1, 5, 6)

Ans. A

Sol. On solving $Ax = C$, & $Bx = D$

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad x = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$$

$$P = (1, 2, 3), \quad Q = (3, 1, 2)$$

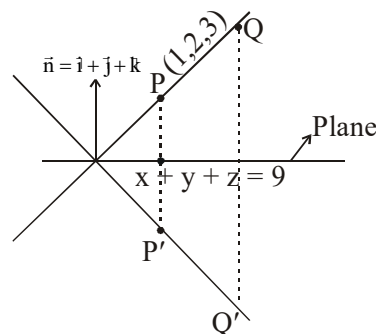
$$PP' ; \frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{1} = \lambda$$

$$(\lambda + 1, \lambda + 2, \lambda + 3) \text{ Lies on plane}$$

$$3\lambda + 6 = 9 \Rightarrow \lambda = 1$$

$$\therefore P' = (3, 4, 5)$$

$$\text{similarly } Q' = (5, 3, 4)$$



V01/1/041. $\hat{a}$, $\hat{b}$ and $\hat{a} - \hat{b}$ are unit vectors. The volume of the parallelopiped, formed with $\hat{a}$, $\hat{b}$ and $\hat{a} \times \hat{b}$ as cotermious edges is :

- (A) 1 (B) $\frac{1}{4}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

Ans. D

Sol. $|\hat{a} - \hat{b}| = 1$

$$\Rightarrow \cos\theta = \frac{1}{2}$$

$$\begin{aligned}\text{Volume of parallelopiped} &= [\hat{a} \hat{b} \hat{a} \times \hat{b}] \\ &= \sin^2 \theta = \frac{3}{4}\end{aligned}$$

V01/1/042. If $\vec{a}, \vec{b}$ and $\vec{c}$ are three non-zero non-coplanar vectors and $\vec{p} = \vec{a} + \vec{b} - 2\vec{c}$; $\vec{q} = 3\vec{a} - 2\vec{b} + \vec{c}$ and $\vec{r} = \vec{a} - 4\vec{b} + 2\vec{c}$ are three vectors such that the volumes of the parallelopiped formed by $\vec{a}, \vec{b}, \vec{c}$ and $\vec{p}, \vec{q}, \vec{r}$ as their coterminal edges are V_1 and V_2 respectively. Then $\frac{V_2}{V_1}$ is equal to :

- (A) 10 (B) 15 (C) 20 (D) None of these

Ans. B

Sol. $V_1 = [\vec{a} \quad \vec{b} \quad \vec{c}]$

$$\begin{aligned}V_2 &= [\vec{a} + \vec{b} - 2\vec{c} \quad 3\vec{a} - 2\vec{b} + \vec{c} \quad \vec{a} - 4\vec{b} + 2\vec{c}] \\ &= 15[\vec{a} \quad \vec{b} \quad \vec{c}]\end{aligned}$$

V01/1/043. The distance between the line $\vec{r} = 2\hat{i} - 2\hat{j} + 3\hat{k} + \lambda(\hat{i} - \hat{j} + 4\hat{k})$ and the plane $\vec{r} \cdot (\hat{i} + 5\hat{j} + \hat{k}) = 5$ is :

- (A) $\frac{10}{9}$ (B) $\frac{10}{3\sqrt{3}}$ (C) $\frac{3}{10}$ (D) $\frac{10}{3}$

Ans. B

Sol. $\vec{p} \cdot \vec{n} = 0$

$\Rightarrow$ Line is parallel to plane

$\Rightarrow$ distance = perpendicular distance of $(2, -2, 3)$ from plane

$$= \frac{|2 - 10 + 3 - 5|}{\sqrt{1 + 25 + 1}} = \frac{10}{3\sqrt{3}}$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

V01/2/044. If $\vec{q}, \vec{r}$ are unit vectors such that $\vec{p} = \vec{q} \times \vec{p} + \vec{r}$, then :

- (A) $\vec{p} \cdot \vec{q} = \vec{r} \cdot \vec{q}$ (B) maximum value of $[\vec{p} \vec{q} \vec{r}] = \frac{1}{2}$

(C) minimum value of $[\vec{p} \vec{q} \vec{r}] = 0$

(D) minimum value of $[\vec{p} \vec{q} \vec{r}] = -1$

Ans. ABC

Sol. Take dot product with $\vec{q}$ both the sides

$$\vec{p} \cdot \vec{q} = \vec{r} \cdot \vec{p}$$

$$\vec{p} - \vec{q} \times \vec{p} = \vec{r}$$

$$\text{Squaring } p^2 + p^2 \sin^2 \theta = 1$$

$$p^2 = \frac{1}{1 + \sin^2 \theta}$$

$$\vec{p} = \vec{q} \times \vec{p} + \vec{r}$$

$$\text{Squaring } p^2 = p^2 \sin^2 \theta + 2[\vec{q} \vec{p} \vec{r}] + 1$$

$$2[\vec{p} \vec{q} \vec{r}] = 1 - p^2 \cos^2 \theta = 1 - \frac{\cos^2 \theta}{(1 + \sin^2 \theta)}$$

$$2[\vec{p} \vec{q} \vec{r}] = 1 - \frac{(1 - \sin^2 \theta)}{(1 + \sin^2 \theta)} = \frac{2 \sin^2 \theta}{(1 + \sin^2 \theta)}$$

$$2[\vec{p} \vec{q} \vec{r}] = 2 \left(1 - \frac{1}{1 + \sin^2 \theta} \right)$$

$$[\vec{p} \vec{q} \vec{r}] = 1 - \frac{1}{(1 + \sin^2 \theta)}$$

$$\text{When } \sin^2 \theta = 0 \Rightarrow [\vec{p} \vec{q} \vec{r}] = 0$$

$$\text{When } \sin^2 \theta = 1 \Rightarrow [\vec{p} \vec{q} \vec{r}] = \frac{1}{2}$$

V01/2/045. The plane $x + y + z = 4$ cuts the coordinate axes at A, B and C. Let M be the centre of tetrahedron OABC. The mid points of AM, BM and CM be D, E, & F respectively. If the shortest distance between the line passing through O & E and the line passing through D & F is $\left(\frac{a}{b\sqrt{6}} \right)$ where a & b are coprime numbers, then:

(A) $a + b = 17$

(B) $a - b = 11$

(C) $a + b = 18$

(D) $a - b = 12$

Ans. AB

Sol. A(4,0,0), B(0,4,0), C(0,0,4)
M(1,1,1)

$$D\left(\frac{5}{2}, \frac{1}{2}, \frac{1}{2}\right), E\left(\frac{1}{2}, \frac{5}{2}, \frac{1}{2}\right), F\left(\frac{1}{2}, \frac{1}{2}, \frac{5}{2}\right)$$

$$\text{Line } \overrightarrow{OE} \text{ is } \lambda \left(\frac{1}{2}\hat{i} + \frac{5}{2}\hat{j} + \frac{1}{2}\hat{k} \right)$$

$$\& \text{ line } \overrightarrow{DF} \text{ is } \left(\frac{5}{2}\hat{i} + \frac{1}{2}\hat{j} + \frac{1}{2}\hat{k} \right) + \mu(-2\hat{i} + 2\hat{k})$$

$$\therefore \text{ Shortest distance } d = \frac{|(\vec{a}_1 - \vec{a}_2)(\vec{b}_1 \times \vec{b}_2)|}{|\vec{b}_1 \times \vec{b}_2|}$$

$$= \frac{\left| \left(\frac{5}{2}\hat{i} + \frac{1}{2}\hat{j} + \frac{1}{2}\hat{k} \right) \cdot (5\hat{i} - 2\hat{j} + 5\hat{k}) \right|}{3\sqrt{6}}$$

$$= \frac{|25 - 2 + 5|}{6\sqrt{6}} = \frac{14}{3\sqrt{6}}$$

V01/2/046. Let $\vec{a} = \hat{i} + \hat{j} - \hat{k}$, $\vec{b} = \hat{i} - \hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j}$. If $\vec{d}$ is a unit vector such that $\vec{a} \cdot \vec{d} = 0$ and $[\vec{b} \ \vec{c} \ \vec{d}] = 0$, then $\vec{d}$ equals :

$$(A) \frac{3\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{14}}$$

$$(B) \frac{\hat{i} + 2\hat{j} + 3\hat{k}}{\sqrt{14}}$$

$$(C) \frac{-\hat{i} - 2\hat{j} + 3\hat{k}}{\sqrt{14}}$$

$$(D) \left(\frac{-3\hat{i} + 2\hat{j} - \hat{k}}{\sqrt{14}} \right)$$

Ans. AD

Sol. Let $\vec{d} = d_1\hat{i} + d_2\hat{j} + d_3\hat{k}$

$$[\vec{b} \ \vec{c} \ \vec{d}] = 0 \Rightarrow (\vec{b} \times \vec{c}) \cdot \vec{d} = 0$$

$$\Rightarrow (-\hat{i} - 2\hat{j} - \hat{k}) \cdot \vec{d} = 0$$

$$\Rightarrow d_1 + 2d_2 + d_3 = 0 \quad \dots(1)$$

$$\text{Again } \vec{a} \cdot \vec{d} = 0$$

$$\Rightarrow d_1 + d_2 - d_3 = 0 \quad \Rightarrow d_1 + d_2 + d_3 \quad \dots(2)$$

$$\text{and } d_1^2 + d_2^2 + d_3^2 = 1 \quad \dots(3)$$

from (1) and (2)

$$d_2 + 2d_3 = 0$$

$$d_2 = -2d_3$$

$$d_1 = 3d_3$$

$$9d_3^2 + 4d_3^2 + d_3^2 = 1$$

$$d_3 = \pm \frac{1}{\sqrt{14}}$$

$$\text{Hence } \vec{d} = \frac{3\hat{i} - 2\hat{j} + \hat{k}}{\sqrt{14}}$$

$$\text{or } \vec{d} = \frac{-3\hat{i} + 2\hat{j} - \hat{k}}{\sqrt{14}}$$

V01/2/047. Let $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$. If $\vec{c}$ be a unit vector such that $[\vec{a} \ \vec{b} \ \vec{c}]$ takes its maximum value M, then:

$$(A) \ M = 3\sqrt{3} \quad (B) \ M = 2\sqrt{3} \quad (C) \ \vec{c} = \frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}} \quad (D) \ \vec{c} = \frac{-\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$$

Ans. AC

$$\text{Sol. } \because [\vec{a} \ \vec{b} \ \vec{c}] = |\vec{a}| |\vec{b}| |\vec{c}| \sin \theta \cos \phi, \theta = \vec{a} \wedge \vec{b}, \phi = (\vec{a} \times \vec{b}) \wedge \vec{c}$$

$$= |\vec{a}| |\vec{b}| \sin \theta \cos \phi \leq |\vec{a}| |\vec{b}| \sin \theta \leq |\vec{a} \times \vec{b}| \quad \left(\because |\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta |\hat{n}| \right)$$

$$\Rightarrow [\vec{a} \ \vec{b} \ \vec{c}]_{\max} = |\vec{a} \times \vec{b}| \text{ when } \phi = 0$$

$\because$ angle between $\vec{c}$ and $\vec{a} \times \vec{b}$ is 0

$$\Rightarrow \vec{c} = \lambda (\vec{a} \times \vec{b}), \lambda > 0 \text{ (right handed system)}$$

$\because \vec{c}$ is unit vector

$$\therefore \vec{c} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$$

$$\therefore [\vec{a} \ \vec{b} \ \vec{c}]_{\max} = |\vec{a} \times \vec{b}| = 3\sqrt{3} = M$$

$$\text{and } \vec{c} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$$

V01/2/048. Which of the following is/are true :

(A) any four non-zero vectors in space will always be linearly independent.

(B) area of parallelogram with diagonals $\vec{d}_1 = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{d}_2 = -\hat{i} + 2\hat{j} - 5\hat{k}$ is equal to $2\sqrt{5}$ sq. units

(C) $\begin{bmatrix} \vec{a} \times \vec{b} & \vec{b} \times \vec{c} & \vec{a} \end{bmatrix} = (\vec{a} \cdot \vec{b}) \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix}$

(D) projection of $\hat{i} + \hat{j} + \hat{k}$ on $-2\hat{i} + 2\hat{j} + \hat{k}$ is $\frac{1}{3}$

Ans. BCD

Sol. (A) very basic, do yourself

(B) $A = \frac{1}{2} |\vec{d}_1 \times \vec{d}_2| = 2\sqrt{5}$

(C) $\begin{bmatrix} \vec{a} \times \vec{b} & \vec{b} \times \vec{c} & \vec{a} \end{bmatrix} = (\vec{a} \times \vec{b}) \cdot ((\vec{b} \times \vec{c}) \times \vec{a})$
 $= (\vec{a} \times \vec{b}) \cdot ((\vec{a} \cdot \vec{b})\vec{c} - (\vec{a} \cdot \vec{c})\vec{b})$
 $= ((\vec{a} \times \vec{b}) \cdot \vec{c})(\vec{a} \cdot \vec{b}) - ((\vec{a} \times \vec{b}) \cdot \vec{b})(\vec{a} \cdot \vec{c})$
 $= (\vec{a} \cdot \vec{b})[\vec{a} \vec{b} \vec{c}] - 0 = (\vec{a} \cdot \vec{b})[\vec{a} \vec{b} \vec{c}]$

(D) $(\hat{i} + \hat{j} + \hat{k}) \cdot \frac{(-2\hat{i} + 2\hat{j} + \hat{k})}{3}$
 $= \frac{-2 + 2 + 1}{3} = \frac{1}{3}$

V01/2/049. Let $\vec{a}, \vec{b}, \vec{c}$ are three non coplanar vectors such that $\vec{\alpha} = \vec{a} + 2\vec{b} + 3\vec{c}, \vec{\beta} = \vec{a} + 3\vec{b} + 4\vec{c}$ and $\vec{\gamma} = \vec{a} + \vec{b} + \vec{c}$ and $\vec{\lambda} = 21\vec{a} + 41\vec{b} + 54\vec{c}$. If $\vec{\lambda} = x_1\vec{\alpha} + x_2\vec{\beta} + x_3\vec{\gamma}$, then :

(A) $x_3 > x_2 > x_1$

(B) x_1, x_2, x_3 are in A.P.

(C) H.C.F. of x_1 & x_3 is 2

(D) x_1, x_2, x_3 are in H.P.

Ans. ABC

Sol. $21\vec{a} + 41\vec{b} + 54\vec{c} = \vec{a}(x_1 + x_2 + x_3) + \vec{b}(2x_1 + 3x_2 + x_3) + \vec{c}(3x_1 + 4x_2 + x_3)$

By comparison $x_1 + x_2 + x_3 = 21$

$2x_1 + 3x_2 + x_3 = 41$

$3x_1 + 4x_2 + x_3 = 54$

By solving $x_1 = 6, x_2 = 7, x_3 = 8$

V01/2/050. Let $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $|\vec{a}| = |\vec{b}| = |\vec{c}| = 2$ and $\vec{a} \wedge \vec{b} = \vec{b} \wedge \vec{c} = \vec{c} \wedge \vec{a} = \frac{\pi}{3}$, then :

(A) volume of parallelopiped formed by $\vec{a}, \vec{b}, \vec{c}$ as three coterminuous edges is $8\sqrt{2}$

(B) volume of parallelopiped formed by $\vec{a}, \vec{b}, \vec{c}$ as three coterminuous edges is $4\sqrt{2}$

(C) height of parallelopiped formed by $\vec{a}, \vec{b}, \vec{c}$ as three coterminuous edges is $\frac{2\sqrt{2}}{\sqrt{3}}$ whose base is a parallelogram with adjacent sides $\vec{a}, \vec{b}$.

(D) height of parallelopiped formed by $\vec{a}, \vec{b}, \vec{c}$ as three coterminuous edges is $\frac{4\sqrt{2}}{\sqrt{3}}$ whose base is a parallelogram with adjacent sides $\vec{a}, \vec{b}$.

Ans. BC

Sol.
$$[\vec{a} \ \vec{b} \ \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

$$[\vec{a} \ \vec{b} \ \vec{c}]^2 = \begin{vmatrix} 4 & 2 & 2 \\ 2 & 4 & 2 \\ 2 & 2 & 4 \end{vmatrix} = 4(12) - 2(4) + 2(-4) = 32$$

$$v = 4\sqrt{2}$$

$$v = (\text{base area}) (\text{ht}) \Rightarrow 4\sqrt{2} = 4 \times \frac{\sqrt{3}}{2} \times \text{ht}$$

$$\therefore \text{base area} = \text{area of parallogram} = |\vec{a} \times \vec{b}| = 2 \cdot 2 \cdot \frac{\sqrt{3}}{2}$$

$$\Rightarrow \frac{2\sqrt{2}}{\sqrt{3}} = \text{ht}$$

V01/2/051. Consider a pyramid P-ABCD whose base ABCD is a square and whose vertex P is equidistant from A, B, C and D. If AB = 1 and $\angle APD = 60^\circ$, then :

(A) Volume of pyramid is $\frac{1}{3\sqrt{2}}$

(B) Shortest distance between the skew lines AP and BD is $\frac{1}{\sqrt{3}}$

(C) If P' be the image of P with respect to plane ABCD then $|\overline{PP'}| = \sqrt{2}$

(D) Volume of pyramid is $\frac{1}{6\sqrt{2}}$

Ans.

AC

Sol.

Using $\triangle PMA$

$$PM = \frac{1}{2} \cot 30^\circ = \frac{\sqrt{3}}{2}$$

using $\triangle POM$

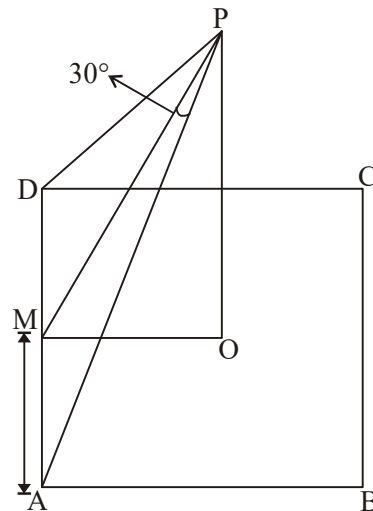
$$OP = \sqrt{\frac{3}{4} - \frac{1}{4}} = \frac{1}{\sqrt{2}}$$

$\therefore$ volume of pyramid

$$= \frac{1}{3} \cdot \text{area of base} \times \text{height} = \frac{1}{3} \cdot 1^2 \cdot \frac{1}{\sqrt{2}}$$

$$= \frac{1}{3\sqrt{2}}$$

and $|\overline{PP'}| = 2|\overline{OP}| = \sqrt{2}$



V01/2/052. A, B & C be three points in space such that $A \equiv (3, 4, 2)$, $B \equiv (3, 0, 2)$ & $C \equiv (5, 0, 0)$, then :

(A) distance of orthocentre of $\triangle ABC$ from origin is $\sqrt{13}$

(B) the position vector of circumcentre of $\triangle ABC$ is $4\hat{i} + 2\hat{j} + \hat{k}$

(C) if point D forms parallelogram with remaining three points, then $D \equiv (5, 4, 0)$

(D) If point D forms parallelogram with remaining three points then $D \equiv (5, -4, 0)$

Ans.

ABCD

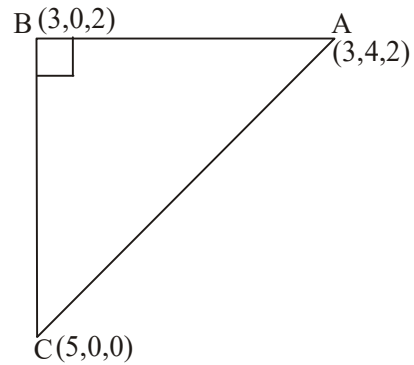
Sol. $AB \perp BC$
 $\Rightarrow \triangle ABC$ right angle at B
 $\therefore B$ is orthocentre

(A) $OB = \sqrt{13}$

(B) Circumcentre is mid-point of AC i.e. (4,2,1)

(C) If $D \equiv (5, 4, 0)$ mid-point of BD & AC is same.

(D) If $D \equiv (5, -4, 0)$ mid-point of AD & BC is same.



V01/2/053. Consider points $A(0,0,0)$, $B(1,1,1)$ and plane $P: x - y + z = 3$. Let $Q(a,b,c)$ be a moving point on plane P then identify the correct statement(s):

(A) If $QA + QB$ is minimum then $a - c - 10b$ is equal to 1

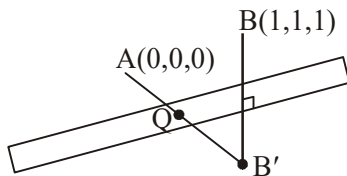
(B) If $QA + QB$ is minimum then $a - c - 10b$ is equal to 2

(C) If $|QA - QB|$ is maximum then $a - 2b + c$ is equal to 0

(D) If $|QA - QB|$ is minimum then Q is unique

Ans. BC

Sol.



For $(QA + QB)$ min, Q must lie on AB' line where B' is mirror image of B in plane $x - y + z = 3$

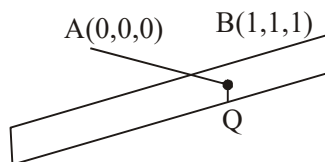
$$B' \equiv \left(\frac{7}{3}, \frac{-1}{3}, \frac{7}{3} \right)$$

Q is intersection of AB' and plane

$$Q \left(\frac{7}{5}, \frac{-1}{5}, \frac{7}{5} \right) \equiv (a, b, c)$$

Hence $a - c - 10b = 2$

For $|QA - QB|_{\max}$, Q must lie on point of intersection of AB with plane



$$Q \equiv (3, 3, 3) \equiv (a, b, c)$$

V01/2/054. A plane P contains the line $L_1: \frac{y}{b} + \frac{z}{c} = 1, x = 0$ and is parallel to the line $L_2: \frac{x}{a} - \frac{z}{c} = 1, y = 0$, then:

(A) equation of plane Π is $-\frac{x}{a} + \frac{y}{b} + \frac{z}{c} - 1 = 0$

(B) equation of plane Π is $\frac{x}{a} + \frac{y}{b} - \frac{z}{c} - 1 = 0$

(C) if shortest distance between line L_1 and line L_2 is $\frac{1}{4}$, then $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$ is 64

(D) if shortest distance between line L_1 and line L_2 is $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$, then is 192

Ans. AC

Sol. Equation of $L_1 \equiv \frac{x}{0} = \frac{y}{-b} = \frac{z-c}{c}$

Equation of $L_2 \equiv \frac{x}{a} = \frac{y}{0} = \frac{z+c}{c}$

Direction of Π is along the vector $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -b & c \\ a & 0 & c \end{vmatrix}$

$$= -bc\hat{i} + ac\hat{j} + ab\hat{k}$$

Equation of Π is

$$-bc(x-0) + ac(y-0) + ab(z-c) = 0$$

$$-\frac{x}{a} + \frac{y}{b} + \frac{z}{c} - 1 = 0$$

Distance between L_1 & L_2 is

$$= \frac{|(0, 0, 2c) \cdot (bc, -ac, -ab)|}{\sqrt{b^2c^2 + a^2c^2 + a^2b^2}}$$

$$\therefore \frac{1}{4} = \frac{|2abc|}{\sqrt{b^2c^2 + c^2a^2 + a^2b^2}}$$

$$\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} = 64$$

V01/2/055. The plane $ax + 4y + z = 0$, $2y + 3z - 1 = 0$, $3x - bz + 2 = 0$ must :

(A) meet at a point if $ab \neq 15$

(B) meet on a line if $ab = 15$, $a = 3$

(C) have no common point if $ab = 15$, $a \neq 3$

(D) have no common point if $ab = 15$, $b \neq 5$

Ans. ABCD

Sol. $ax + 4y + z = 0$

$$0x + 2y + 3z = 1$$

$$3x + 0y - bz = -2$$

$$\Delta = \begin{vmatrix} a & 4 & 1 \\ 0 & 2 & 3 \\ 3 & 0 & -b \end{vmatrix} = 2(15 - ab)$$

$$\Delta_1 = \begin{vmatrix} 0 & 4 & 1 \\ 1 & 2 & 3 \\ -2 & 0 & -b \end{vmatrix} = 4(b - 5)$$

$$\Delta_2 = \begin{vmatrix} a & 0 & 1 \\ 0 & 1 & 3 \\ 3 & -2 & -b \end{vmatrix} = -ab + 6a - 3$$

$$\Delta_3 = \begin{vmatrix} a & 4 & 0 \\ 0 & 2 & 1 \\ 3 & 0 & -2 \end{vmatrix} = 4(3 - a)$$

Unique solution if $ab \neq 15$ (point)

Infinite solution if $ab = 15$ & $a = 3$ & $b = 5$ (line)

No solution if $ab = 15$ & ($a \neq 3$ or $b \neq 5$)

V01/2/056. The volume of the parallelopiped whose coterminous edges are represented by the vectors $2\vec{b} \times \vec{c}$, $3\vec{c} \times \vec{a}$

and $4\vec{a} \times \vec{b}$ where $\vec{a} = (1 + \sin \theta)\hat{j} + \sin 2\theta\hat{k}$, $\vec{b} = \sin\left(\theta + \frac{2\pi}{3}\right)\hat{i} + \cos\left(\theta + \frac{2\pi}{3}\right)\hat{j} + \sin\left(2\theta + \frac{4\pi}{3}\right)\hat{k}$,

$\vec{c} = \sin\left(\theta - \frac{2\pi}{3}\right)\hat{i} + \cos\left(\theta - \frac{2\pi}{3}\right)\hat{j} + \sin\left(2\theta - \frac{4\pi}{3}\right)\hat{k}$ is 18 cubic units, then the values of θ , in the interval

$\left(0, \frac{\pi}{2}\right)$; is/are :

(A) $\frac{\pi}{9}$

(B) $\frac{2\pi}{9}$

(C) $\frac{\pi}{3}$

(D) $\frac{4\pi}{9}$

Ans. ABD

Sol.

$$\text{Volume} = \begin{vmatrix} 4\vec{a} \times \vec{b} & 2\vec{b} \times \vec{c} & 3\vec{c} \times \vec{a} \end{vmatrix} = 18$$

$$24 \begin{vmatrix} \vec{a} & \vec{b} & \vec{c} \end{vmatrix} = 18$$

$$\begin{vmatrix} 0 & 1 + \sin \theta & \sin 2\theta \\ \sin \left(\theta + \frac{2\pi}{3} \right) & \cos \left(\theta + \frac{2\pi}{3} \right) & \sin \left(2\theta + \frac{4\pi}{3} \right) \\ \sin \left(\theta - \frac{2\pi}{3} \right) & \cos \left(\theta - \frac{2\pi}{3} \right) & \sin \left(2\theta - \frac{4\pi}{3} \right) \end{vmatrix} = \frac{3}{4}$$

$$-(1 + \sin \theta) \left[\sin \left(\theta + \frac{2\pi}{3} \right) \sin \left(2\theta - \frac{4\pi}{3} \right) - \sin \left(2\theta + \frac{4\pi}{3} \right) \sin \left(\theta - \frac{2\pi}{3} \right) \right]$$

$$+ \sin 2\theta \left[\sin \left(\left(\theta + \frac{2\pi}{3} \right) - \left(\theta - \frac{2\pi}{3} \right) \right) \right] = \pm \frac{\sqrt{3}}{2}$$

$$\frac{(1 + \sin \theta)}{2} \left[\cos \left(3\theta - \frac{2\pi}{3} \right) - \cos (2\pi - \theta) - \cos \left(3\theta + \frac{2\pi}{3} \right) + \cos (\theta - 2\pi) \right]$$

$$+ \sin 2\theta \left(\frac{-\sqrt{3}}{2} \right) = \pm \frac{\sqrt{3}}{2}$$

$$\frac{1 + \sin \theta}{2} \cdot 2 \sin 3\theta \sin \frac{2\pi}{3} + (\sin 2\theta) \left(\frac{-\sqrt{3}}{2} \right) = \pm \frac{\sqrt{3}}{2}$$

$$\sin 3\theta (1 + \sin \theta) - \sin 2\theta = \pm 1$$

V01/2/057. Let O, P, Q, R, S (O being origin) be any five points lying in a plane in order. P, Q, R, S have respective position vectors $\vec{p}, \vec{q}, \vec{r}, \vec{s}$ and let ΔXYZ and $d(X, Y)$ represent respectively the area of the triangle XYZ and the distance between points X and Y, then

(A) $(\vec{p} \times \vec{q}) \cdot (\vec{r} \times \vec{s}) + (\vec{p} \times \vec{s}) \cdot (\vec{q} \times \vec{r}) = (\vec{p} \times \vec{r}) \cdot (\vec{q} \times \vec{s})$

(B) $\Delta POQ \cdot \Delta ROS + \Delta POS \cdot \Delta QOR > \Delta POR \cdot \Delta QOS$

(C) $\sin \angle POQ \cdot \sin \angle ROS + \sin \angle POS \cdot \sin \angle QOR = \sin \angle POR \cdot \sin \angle QOS$

(D) If the five points are taken on a circle then $d(P, Q) \cdot d(R, S) + d(P, S) \cdot d(Q, R) = d(P, R) \cdot d(Q, S)$

Ans. ACD

Sol. using the identity $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c})$, we observe that

$$(\vec{p} \times \vec{q}) \cdot (\vec{r} \times \vec{s}) + (\vec{p} \times \vec{s}) \cdot (\vec{q} \times \vec{r}) = (\vec{p} \times \vec{r}) \cdot (\vec{q} \times \vec{s})$$

$$\Rightarrow \sin \angle POQ \cdot \sin \angle ROS + \sin \angle POS \cdot \sin \angle QOR = \sin \angle POR \cdot \sin \angle QOS$$

Now multiply both sides by the square of circumdiameter.

V01/2/058. Let P be a point not on the line L that passes through the points Q and R where $\overrightarrow{QR} = \vec{a}$ and $\overrightarrow{QP} = \vec{b}$. The distance d from the point P to the line L, is equal to :

(A) $\frac{|\vec{a} \times \vec{b}|}{|\vec{b}|}$ (B) $\frac{|\vec{a} \times \vec{b}|}{|\vec{a}|}$ (C) $\left| \vec{b} - \left(\frac{\vec{a} \cdot \vec{b}}{\vec{a} \cdot \vec{a}} \right) \vec{a} \right|$ (D) $\sqrt{|\vec{b}|^2 - \left(\frac{\vec{b} \cdot \vec{a}}{|\vec{a}|} \right)^2}$

Ans. BCD

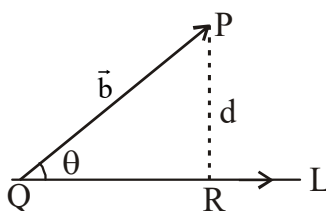
Sol. $\sin \theta = \frac{d}{|\vec{b}|}$

$$d = |\vec{b}| \cdot \sin \theta$$

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$$

$$|\vec{a} \times \vec{b}| = |\vec{a}| \times d$$

$$d = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}|}$$



V01/2/059. Let $\vec{a} = \hat{i} - \hat{j}$, $\vec{b} = -\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c} = \hat{i} - \hat{k}$, then the value of $\left| (\vec{a} \times \vec{b}) \times \vec{c} \right| + \left| (\vec{a} + \vec{b} + \vec{c}) \right| - \sqrt{2} |\vec{b} \times \vec{c}|$ is equal to :

(A) the value of definite integral $\int_0^{\pi} \cos^3 x \, dx$

(B) the value of $[\vec{a} \ \vec{b} \ \vec{c}]$

(C) Number of solutions of the equation $||x| - 1| = e^{-x}$

(D) $\lim_{x \rightarrow \infty} \frac{x+1}{e^{2x} + e^x + 2}$

Ans. ABD

Sol. $\vec{a} = \hat{i} - \hat{j}$, $\vec{b} = -\hat{i} + 2\hat{j} - \hat{k}$, $\vec{c} = \hat{i} - \hat{k}$

$$\vec{a} + \vec{b} + \vec{c} = \hat{i} + \hat{j} - 2\hat{k}$$

$$|\vec{a} + \vec{b} + \vec{c}| = \sqrt{1+1+4} = \sqrt{6}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 0 \\ -1 & 2 & -1 \end{vmatrix}$$

$$= \hat{i}(1-0) - \hat{j}(-1-0) + \hat{k}(2-1)$$

$$= \hat{i} + \hat{j} + \hat{k}$$

$$(\vec{a} \times \vec{b}) \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 1 & 0 & -1 \end{vmatrix}$$

$$= \hat{i}(-1-0) - \hat{j}(-1-1) + \hat{k}(0-1)$$

$$= -\hat{i} + 2\hat{j} - \hat{k}$$

$$\Rightarrow |(\vec{a} \times \vec{b}) \times \vec{c}| = \sqrt{1+4+1} = \sqrt{6}$$

$$\vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & -1 \\ 1 & 0 & -1 \end{vmatrix}$$

$$= \hat{i}(-2-0) - \hat{j}(1+1) + \hat{k}(0-2)$$

$$= -2\hat{i} - 2\hat{j} - 2\hat{k}$$

$$|\vec{b} \times \vec{c}| = \sqrt{4+4+4} = 2\sqrt{3}$$

$$\text{then } |(\vec{a} \times \vec{b}) \times \vec{c}| + |\vec{a} + \vec{b} + \vec{c}| - \sqrt{2} |\vec{b} \times \vec{c}|$$

$$= \sqrt{6} + \sqrt{6} - \sqrt{2} \times 2\sqrt{3}$$

$$= 2\sqrt{6} - 2\sqrt{6} = 0$$

$$(A) \because \cos 3x = 4\cos^3 x - 3\cos x$$

$$\cos^3 x = \frac{\cos 3x + 3\cos x}{4}$$

$$\therefore \int_0^{\pi} \cos^3 x = \frac{1}{4} \int_0^{\pi} \cos 3x \cdot dx + \frac{3}{4} \int_0^{\pi} \cos x \cdot dx$$

$$= \frac{1}{4} \left[\frac{\sin 3x}{3} \right]_0^\pi + 3 [\cos x]_0^\pi = 0$$

$$\begin{aligned} \text{(B)} \quad [\vec{a} \vec{b} \vec{c}] &= \vec{a} \cdot (\vec{b} \times \vec{c}) \\ &= (\hat{i} - \hat{j}) \cdot (-2\hat{i} - 2\hat{j} - 2\hat{k}) \\ &= -2 + 2 = 0 \end{aligned}$$

$$\text{(D)} \quad \lim_{x \rightarrow \infty} \frac{x+1}{e^{2x} + e^x + 2} \quad \left(\frac{\infty}{\infty} \text{ form} \right)$$

L'Hospital rule

$$\lim_{x \rightarrow \infty} \frac{1}{2e^{2x} + e^x} = \lim_{x \rightarrow \infty} \frac{1}{e^x (2e^x + 1)} = 0$$

V01/2/060. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are non-zero, non-collinear vectors such that a vector $\vec{p} = a b \cos(2\pi - (\vec{a} \wedge \vec{b}))\vec{c}$ and a vector $\vec{q} = a c \cos(\pi - (\vec{a} \wedge \vec{c}))\vec{b}$ then $\vec{p} + \vec{q}$ is :

- | | |
|---|---|
| (A) parallel to $\vec{a}$ | (B) perpendicular to $\vec{a}$ |
| (C) coplanar with $\vec{b}$ and $\vec{c}$ | (D) coplanar with $\vec{a}$ and $\vec{c}$ |

Ans. BC

Sol. $\vec{p} = ab \cos(2\pi - (\vec{a} \wedge \vec{b})) \cdot \vec{c} = (\vec{a} \times \vec{b}) \cdot \vec{c}$

$$\vec{q} = ac \cos(\pi - (\vec{a} \wedge \vec{c})) \cdot \vec{b}$$

$$\vec{q} = (\vec{a} \times \vec{c}) \cdot \vec{b}$$

Now

$$\vec{a} \cdot (\vec{p} + \vec{q}) = \vec{a} \cdot \vec{p} + \vec{a} \cdot \vec{q}$$

$$= \vec{a} \cdot (\vec{a} \times \vec{b}) \cdot \vec{c} + \vec{a} \cdot (\vec{a} \times \vec{c}) \cdot \vec{b}$$

$$= [\vec{a} \vec{a} \vec{b}] \cdot \vec{c} + [\vec{a} \vec{a} \vec{c}] \cdot \vec{b}$$

$$\text{but } [\vec{a} \vec{a} \vec{b}] = [\vec{a} \vec{a} \vec{c}] = 0 \\ = 0 + 0 = 0$$

V01/2/061. Consider the lines $L_1 : \frac{x-2}{1} = \frac{y-1}{7} = \frac{z+2}{-5}$ $L_2 : x-4 = y+3 = -z$ Then which of the following is/are correct :

- (A) Point of intersection of L_1 and L_2 is $(1, -6, 3)$
 (B) Equation of plane containing L_1 and L_2 is $x + 2y + 3z + 2 = 0$
 (C) Acute angle between L_1 and L_2 is $\cot^{-1} \left(\frac{13}{15} \right)$
 (D) Equation of plane containing L_1 and L_2 is $x + 2y + 2z + 3 = 0$

Ans. AB

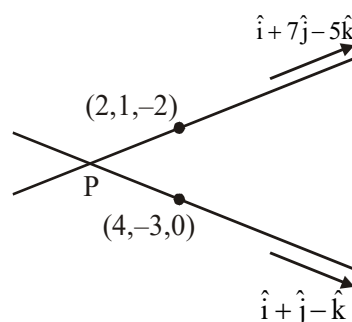
Sol. Point P on line L_1
 $P(2 + \lambda, 1 + 7\lambda, -2 - 5\lambda)$

Point P on line L_2
 $P(4 + r, -3 + r, -r) \Rightarrow \lambda = -1, r = -3$

Acute angle between L_1 and L_2

$$\cos \theta = \frac{13}{15}$$

Equation of plane containing L_1 and L_2 is $x + 2y + 3z + 2 = 0$



V01/2/062. Let $\hat{a}$, $\hat{b}$ and $\hat{c}$ be three unit vectors such that $\hat{a} = \hat{b} + (\hat{b} \times \hat{c})$ then the possible value(s) of $|\hat{a} + \hat{b} + \hat{c}|^2$ can be :

- (A) 1 (B) 4 (C) 16 (D) 9

Ans. AD

Sol. $\hat{a} = \hat{b} + (\hat{b} \times \hat{c})$

$$\hat{a} \cdot \hat{b} = 1 \text{ and } \hat{a} \cdot \hat{c} = \hat{b} \cdot \hat{c}$$

$$|\hat{a} - \hat{b}| = |\hat{b} \times \hat{c}| \Rightarrow \sin \theta = 0 \left(\because \theta = \hat{b} \cdot \hat{c} \right)$$

$$1 = 1 + \sin^2 \theta$$

$$\sin^2 \theta = 0$$

$$\theta = n\pi$$

$$|\hat{a} + \hat{b} + \hat{c}|^2 = 3 + 2(\hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{c} + \hat{a} \cdot \hat{c}) = 5 + 4(\hat{b} \cdot \hat{c}) \\ = 5 \pm 4$$

$$= 1 \text{ or } 9$$

V01/2/063. If $\hat{i} \times [(\vec{a} - \hat{j}) \times \hat{i}] + \hat{j} \times [(\vec{a} - \hat{k}) \times \hat{j}] + \hat{k} \times [(\vec{a} - \hat{i}) \times \hat{k}] = 0$ and $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$ then :

- (A) $x + y = 1$ (B) $y + z = \frac{1}{2}$
 (C) $x + z = 1$ (D) None of these

Ans. AC

Sol. $\hat{i} \times [(\vec{a} - \hat{j}) \times \hat{i}] + \hat{j} \times [(\vec{a} - \hat{k}) \times \hat{j}] + \hat{k} \times [(\vec{a} - \hat{i}) \times \hat{k}] = 0$

$$2\vec{a} - (\hat{i} + \hat{j} + \hat{k}) = 0 \Rightarrow (2x - 1)\hat{i} + (2y - 1)\hat{j} + (2z - 1)\hat{k} = 0$$

$$\Rightarrow x = y = z = \frac{1}{2}$$

V01/2/064. The value of expression $[\vec{a} \times \vec{b} \quad \vec{c} \times \vec{d} \quad \vec{e} \times \vec{f}]$ is equal to :

- (A) $[\vec{a} \vec{b} \vec{d}][\vec{c} \vec{e} \vec{f}] - [\vec{a} \vec{b} \vec{c}][\vec{d} \vec{e} \vec{f}]$ (B) $[\vec{a} \vec{b} \vec{e}][\vec{f} \vec{c} \vec{d}] - [\vec{a} \vec{b} \vec{f}][\vec{e} \vec{c} \vec{d}]$
 (C) $[\vec{c} \vec{d} \vec{a}][\vec{b} \vec{e} \vec{f}] - [\vec{c} \vec{d} \vec{b}][\vec{a} \vec{e} \vec{f}]$ (D) $[\vec{b} \vec{c} \vec{d}][\vec{a} \vec{e} \vec{f}] - [\vec{b} \vec{c} \vec{f}][\vec{a} \vec{e} \vec{d}]$

Ans. ABC

Sol. $[\vec{a} \times \vec{b} \quad \vec{c} \times \vec{d} \quad \vec{e} \times \vec{f}] = (\vec{a} \times \vec{b}) \cdot ((\vec{c} \times \vec{d}) \times (\vec{e} \times \vec{f})) = (\vec{c} \times \vec{d}) \cdot ((\vec{e} \times \vec{f}) \times (\vec{a} \times \vec{b})) = (\vec{e} \times \vec{f}) \cdot ((\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}))$
 $= (\vec{a} \times \vec{b}) \cdot ((\vec{c} \times \vec{d}) \cdot \vec{f}) \vec{e} - ((\vec{c} \times \vec{d}) \cdot \vec{e}) \times \vec{f} = [\vec{c} \quad \vec{d} \quad \vec{f}][\vec{a} \quad \vec{b} \quad \vec{e}] - [\vec{c} \quad \vec{d} \quad \vec{e}][\vec{a} \quad \vec{b} \quad \vec{f}]$

Similarly solve other 2.

V01/2/065. If $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are the pv's of the points A, B, C and D respectively in three dimensional space and satisfy the relation $3\vec{a} - 2\vec{b} + \vec{c} - 2\vec{d} = 0$, then :

- (A) A, B, C and D are coplanar
 (B) the line joining the points B and D divides the line joining the point A and C in the ratio 2 : 1.
 (C) the line joining the points A and C divides the line joining the points B and D in the ratio 1 : 1
 (D) the four vectors $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ are linearly dependent.

Ans. ACD

Sol. $3(\vec{a} - \vec{b}) + (\vec{b} - \vec{c}) + 2(\vec{c} - \vec{d}) = 0$

$$\frac{\overrightarrow{BC} + 2\overrightarrow{CD}}{1+2} = \overrightarrow{BA}$$

V01/2/066. If OABC is a tetrahedron with equal edges and $\hat{p}, \hat{q}, \hat{r}$ are unit vectors along bisectors of

$\overrightarrow{OA}, \overrightarrow{OB} : \overrightarrow{OB}, \overrightarrow{OC} : \overrightarrow{OC}, \overrightarrow{OA}$ respectively and $\hat{a} = \frac{\overrightarrow{OA}}{|\overrightarrow{OA}|}, \hat{b} = \frac{\overrightarrow{OB}}{|\overrightarrow{OB}|}, \hat{c} = \frac{\overrightarrow{OC}}{|\overrightarrow{OC}|}$, then :

$$(A) \frac{[\hat{a} \hat{b} \hat{c}]}{[\hat{p} \hat{q} \hat{r}]} = \frac{3\sqrt{3}}{2}$$

$$(B) \frac{[\hat{a} + \hat{b} \quad \hat{b} + \hat{c} \quad \hat{c} + \hat{a}]}{[\hat{p} + \hat{q} \quad \hat{q} + \hat{r} \quad \hat{r} + \hat{p}]} = \frac{3\sqrt{3}}{4}$$

$$(C) \frac{[\hat{a} + \hat{b} \quad \hat{b} + \hat{c} \quad \hat{c} + \hat{a}]}{[\hat{p} \hat{q} \hat{r}]} = \frac{3\sqrt{3}}{2}$$

$$(D) \frac{[\hat{a} \hat{b} \hat{c}]}{[\hat{p} + \hat{q} \quad \hat{q} + \hat{r} \quad \hat{r} + \hat{p}]} = \frac{3\sqrt{3}}{4}$$

Ans. AD

Sol.
$$\hat{p} = \frac{\hat{a} + \hat{b}}{|\hat{a} + \hat{b}|} = \frac{\hat{a} + \hat{b}}{\sqrt{1+1+2\cos 60^\circ}} = \frac{\hat{a} + \hat{b}}{\sqrt{3}}$$

$$[\hat{p} \quad \hat{q} \quad \hat{r}] = \begin{bmatrix} \frac{\hat{a} + \hat{b}}{\sqrt{3}} & \frac{\hat{b} + \hat{c}}{\sqrt{3}} & \frac{\hat{c} + \hat{a}}{\sqrt{3}} \end{bmatrix}$$

$$[\hat{p} \quad \hat{q} \quad \hat{r}] = \frac{1}{3\sqrt{3}} \cdot 2[\hat{a} \quad \hat{b} \quad \hat{c}]$$

V01/2/067. Let $\hat{a}$ and $\hat{c}$ are unit vectors and $|\vec{b}| = 4$. If the angle between $\hat{a}$ and $\hat{c}$ is $\cos^{-1}\left(\frac{1}{4}\right)$; then $\vec{b} - 2\hat{c} = \lambda\hat{a}$, then the value of λ can be :

(A) 2

(B) -3

(C) 3

(D) -4

Ans. CD

Sol.
$$\vec{b} = 2\hat{c} + \lambda\hat{a}$$

$$|\vec{b}|^2 = 4 + \lambda^2 + 4\lambda\left(\frac{1}{4}\right) = 16 \Rightarrow \lambda = -4, 3$$

V01/2/068. Consider the line $L_1 : x = y = z$ and the line $L_2 : 2x + y + z - 1 = 0 = 3x + y + 2z - 2$, then :

(A) The shortest distance between the two lines is $\frac{1}{\sqrt{2}}$

(B) The shortest distance between the two lines is $\sqrt{2}$

(C) Plane containing the line L_2 and parallel to line L_1 is $z - x + 1 = 0$

(D) Perpendicular distance of origin from plane containing line L_2 and parallel to line L_1 is $\frac{1}{\sqrt{2}}$

Ans. AD

Sol. $L_1 : x = y = z$

$$L_2 : \frac{x-1}{1} = \frac{y+1}{-1} = \frac{z}{-1}$$

$$\text{Shortest distance} = \frac{1}{\sqrt{2}}$$

Equation of plane containing line L_2 and parallel to L_1 $y - z + 1 = 0$

$$\text{Distance of origin from this plane} = \frac{1}{\sqrt{2}}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question V01/3/069 to V01/3/070

Let $\vec{a}, \vec{b}, \vec{c}$ are non zero vectors such that $\vec{c} \times (\vec{a} \times \vec{b}) = \vec{a} \times (\vec{a} \times \vec{b})$. Also $|\vec{a} \times \vec{b}| = 2$ and $[\vec{a} \ \vec{b} \ \vec{c}] = 8$ and $[\vec{a}] = 3$.

V01/3/069. $|\vec{c}|$ is equal to :

- (A) 3 (B) 5 (C) 6 (D) $\sqrt{15}$

Ans. B

Sol. $(\vec{c} - \vec{a}) \times (\vec{a} \times \vec{b}) = \vec{0}$

$$\Rightarrow \vec{c} - \vec{a} = \lambda (\vec{a} \times \vec{b})$$

$$\Rightarrow \vec{c} = \vec{a} + \lambda (\vec{a} \times \vec{b})$$

taking dot product with $(\vec{a} \times \vec{b})$ on both sides

$$[\vec{a} \ \vec{b} \ \vec{c}] = 0 + \lambda |\vec{a} \times \vec{b}|^2 \Rightarrow \lambda = 2$$

$$\Rightarrow \vec{c} - \vec{a} = 2(\vec{a} \times \vec{b})$$

$$\text{also } |\vec{c}|^2 = |\vec{a}|^2 + 4|\vec{a} \times \vec{b}|^2 + 2\vec{a} \cdot (\vec{a} \times \vec{b})$$

$$|\vec{c}|^2 = 9 + 16 + 0$$

$$|\vec{c}| = 5$$

V01/3/070. $|\vec{c} - \vec{a}|$ is equal to :

- (A) 1 (B) 9 (C) 4 (D) 25

Ans. C

Sol. $|\vec{c} - \vec{a}| = \sqrt{25 + 16 - 2(\vec{a} \cdot \vec{c})}$

$$\sqrt{25 + 16 - 2(9)} = \sqrt{16} = 4$$

Paragraph for Question V01/3/071 to V01/3/072

Let $P_1 : 2x - 3y + 6z + 8 = 0$ and $P_2 : 3x - 2y - 2z + 7 = 0$ be two planes and $A\left(0, \frac{29}{9}, \frac{5}{18}\right)$ lies on both planes. Two points $B(2, 0, -2)$ and C are such that line of intersection of the planes is the internal angle bisector of $\angle A$ of $\triangle ABC$ and $AB = AC$.

On the basis of above information, answer the following questions :

V01/3/071. If co-ordinates of C are (α, β, γ) , then $(\beta + \gamma)$ is equal to :

- (A) 6 (B) 4 (C) 2 (D) 0

Ans. A

Sol. Point B lies one on plane P_1 . So, point C is the mirror image of point B about the plane P_2 .

$$\text{So, } \frac{\alpha - 2}{3} = \frac{\beta}{-2} = \frac{\gamma + 2}{-2} = K \text{ (say)}$$

$$\Rightarrow C(\alpha, \beta, \gamma) \equiv (3K + 2, -2K, -2K - 2)$$

Now, mid-point of BC will lie on P_2 .

$$\frac{9K + 12}{2} + 2K - 2(-K - 2) + 7 = 0$$

$$\Rightarrow K = -2 \Rightarrow C(-4, 4, 2)$$

$$(\beta + \gamma) = 4 + 2 = 6$$

V01/3/072. The x-intercept of plane containing the point C and perpendicular to planes P_1 and P_2 is :

- (A) 0 (B) $\frac{13}{11}$ (C) $\frac{13}{9}$ (D) $\frac{26}{5}$

Ans. C

Sol. Plane

$$\begin{vmatrix} x+4 & y-4 & z-2 \\ 2 & -3 & 6 \\ 3 & -2 & -2 \end{vmatrix} = 0$$

For x intercept

Put $y = z = 0$

$$\begin{vmatrix} x+4 & -4 & -2 \\ 2 & -3 & 6 \\ 3 & -2 & -2 \end{vmatrix} = 0$$

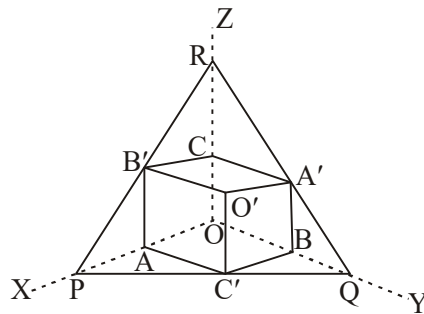
$$(x+4)(18) + 4(-22) - 2(5) = 0$$

$$18x = 26$$

$$x = \frac{13}{9}$$

Paragraph for Question V01/3/073 to V01/3/074

Consider a parallelopiped $OAC'B A'CB'O'$ where $OA = 2$, $OB = 3$ and $OC = 4$. If areas of $\triangle OPQ$, $\triangle OQR$ and $\triangle OPR$ are least (as shown in figure), then



V01/3/073. Volume of tetrahedron $OPQR$, is :

- (A) 24 cubic units (B) 32 cubic units
(C) 48 cubic units (D) 16 cubic units

Ans. B

Sol.

$$\text{Area of } \triangle OPR = \frac{1}{2} (2 + 4 \cot \theta) (4 + 2 \tan \theta)$$

$$= 8 + 8 \cot \theta + 2 \tan \theta$$

minimum value occurs when $8 \cot \theta = 2 \tan \theta$

(use $AM \geq GM$)

$$\Rightarrow \tan \theta = 2$$

$$\Rightarrow AP = 2, CR = 4$$

$\Rightarrow A, C$ are mid points of OP and OR

$\Rightarrow B'$ is midpoint of PR .

and $OP = 4, OR = 8$ similarly $OQ = 6$

For minimum area A', B' and C' must be mid points of QR, PR & PQ respectively.

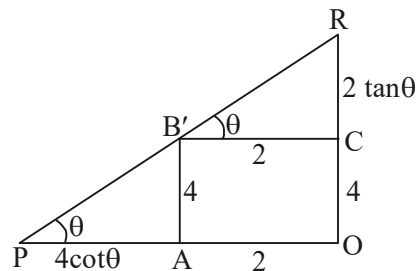
$$\therefore A' (0, 3, 4), Q(0, 6, 0), R(0, 0, 8), C'(2, 3, 0)$$

$$P (4, 0, 0), B (0, 3, 0)$$

$$\text{Volume of tetrahedron} = \frac{1}{6} \left| \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} \right|$$

$$= \frac{1}{6} |\vec{a}| |\vec{b}| |\vec{c}|$$

$$= \frac{1}{6} (4 \times 6 \times 8) = 32$$



V01/3/074. Shortest distance between BA' & PR is :

- (A) 3 (B) $\frac{7}{5}$ (C) $\frac{4}{5}$ (D) $\frac{3}{5}$

Ans. A

Sol.

$$\text{Area of } \triangle OPR = \frac{1}{2} (2 + 4 \cot \theta) (4 + 2 \tan \theta)$$

$$= 8 + 8 \cot \theta + 2 \tan \theta$$

minimum value occurs when $8 \cot \theta = 2 \tan \theta$

(use $AM \geq GM$)

$$\Rightarrow \tan \theta = 2$$

$$\Rightarrow AP = 2, CR = 4$$

$\Rightarrow A, C$ are mid points of OP and OR

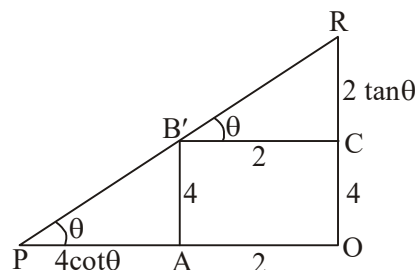
$\Rightarrow B'$ is midpoint of PR .

and $OP = 4, OR = 8$ similarly $OQ = 6$

For minimum area A', B' and C' must be mid points of QR, PR & PQ respectively.

$$\therefore A' (0, 3, 4), Q(0, 6, 0), R(0, 0, 8), C'(2, 3, 0)$$

$$P (4, 0, 0), B (0, 3, 0)$$



Shortest distance is OB i.e. 3

Paragraph for Question V01/3/075 to V01/3/077

Consider a tetrahedron D-ABC with position vectors of its angular points as A(1, 1, 1) ; B(1, 2, 3) and C(1,1,2) Also the position vector of the centre 'G' of the tetrahedron are $\left(\frac{3}{2}\hat{i} + \frac{3}{4}\hat{j} + 2\hat{k}\right)$.

V01/3/075. The volume of the tetrahedron D-ABC is :

- (A) $\frac{2}{3}$ (B) $\frac{1}{3}$ (C) 1 (D) 2

Ans. B

Sol. A(1, 1, 1), B(1, 2, 3), C(1, 1, 2)

D $\equiv$ (3, -1, 2)

$$\overrightarrow{AB} = 0\hat{i} + 1\hat{j} + 2\hat{k}, \overrightarrow{AC} = 0\hat{i} + 0\hat{j} + 1\hat{k}, \overrightarrow{AD} = 2\hat{i} - 2\hat{j} + 1\hat{k}$$

$$\text{Vol}^m = \frac{1}{6} [\overrightarrow{AB} \overrightarrow{AC} \overrightarrow{AD}]$$

$$= \frac{1}{6} \begin{vmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 2 & -2 & 1 \end{vmatrix}$$

$$= \frac{1}{6} [0(0+2) - 1(0-2) + 2(0)]$$

$$= \frac{1}{6} [0+2+0] = \frac{1}{3}$$

V01/3/076. Shortest distance between the skew lines AB and CD is :

- (A) $\sqrt{0.9}$ (B) $\sqrt{0.8}$ (C) 1 (D) $\frac{1}{3}$

Ans. D

Sol. Let D(x₁, y₁, z₁)

$$\text{Centre : } \left(\frac{3}{2}, \frac{3}{4}, 2 \right)$$

$$\left(\frac{1+1+1+x_1}{4}, \frac{1+2+1+y_1}{4}, \frac{1+3+2+z_1}{4} \right) = \left(\frac{3}{2}, \frac{3}{4}, 2 \right)$$

$$\left(\frac{3+x_1}{4}, \frac{4+y_1}{4}, \frac{6+z_1}{4} \right) = \left(\frac{3}{2}, \frac{3}{4}, 2 \right)$$

$$D \equiv (3, -1, 2)$$

$$\text{Eq. of AB : } \vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$$

$$\vec{r} = (\hat{i} + \hat{j} + \hat{k}) + \lambda(\hat{j} + 2\hat{k})$$

$$\text{Eq. of CD : } \vec{r} = (\hat{i} + \hat{j} + 2\hat{k}) + \mu(2\hat{i} - 2\hat{j})$$

$$d = \frac{(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)}{|\vec{b}_1 \times \vec{b}_2|}$$

$$d = \frac{1}{3}$$

V01/3/077. If N is the foot of the perpendicular from the point D on the face ABC then the position vector of N are :

$$(A) \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

$$(B) \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix}$$

$$(C) \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

(D) None of these

Ans. C

Sol. $\because D : (3, -1, 2)$

Eq. of ABC

A(1, 1, 1), B(1, 2, 3), C(1, 1, 2)

Dr. of plane (l, m, n)

$$l(x-1) + m(y-1) + n(z-1) = 0 \quad \text{_____ (1)}$$

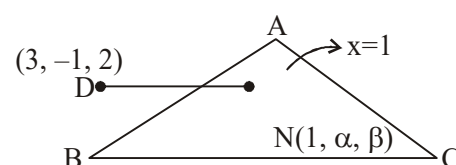
Plane passes through B and C so

$$l(0) + m(2-1) + n(3-1) = 0$$

$$0l + m + 2n = 0 \quad \text{_____ (2)}$$

$$\text{and } l(0) + m(0) + n(2-1) = 0$$

$$0l + 0m + 1n = 0 \quad \text{_____ (3)}$$



$$\Rightarrow \frac{l}{1} = \frac{m}{0} = \frac{n}{0} = k \text{ (let)}$$

$$l = k, m = 0, n = 0$$

$$\text{Equ. } k(x - 1) + 0 + 0 = 0$$

$$x = 1$$

Equ. of plane

$$\text{Drs of DN} \equiv (3 - 1, -1, -\alpha, 2 - \beta) \equiv (\alpha, -1 - \alpha, 2 - \beta)$$

$$\text{DRs of plane AB : } (1, 0, 0)$$

$\therefore$ Angle between plane and line is 90°

$$\frac{2}{1} = \frac{-1 - \alpha}{0} = \frac{2 - \beta}{0}$$

$$\alpha = -1, \beta = 2$$

$$N(1, -1, 2)$$

Paragraph for Question V01/3/078 to V01/3/080

A line L_1 with direction ratios $\langle -3, 2, 4 \rangle$ passes through the point $A(7, 6, 2)$ and a line L_2 with direction ratios $\langle 2, 1, 3 \rangle$ passes through the point $B(5, 3, 4)$. A line L_3 with direction ratios $\langle 2, -2, -1 \rangle$ intersects L_1 and L_2 at C and D .

V01/3/078. The length CD is equal to :

- (A) 4 (B) 6 (C) 9 (D) 11

Ans. C

Sol. For L_1 : DR's : $-3, 2, 4$; pt $(7, 6, 2)$

$$\frac{x-7}{-3} = \frac{y-6}{2} = \frac{z-2}{4} = \lambda \text{ (let)}$$

$$C(-3\lambda + 7, 2\lambda + 6, 4\lambda + 2)$$

for L_2 : DR's : $2, 1, 3$; pt. $(5, 3, 4)$

$$\frac{x-5}{2} = \frac{y-3}{1} = \frac{z-4}{3} = \mu \text{ (let)}$$

$$D(2\mu + 5, \mu + 3, 3\mu + 4)$$

for L_3 : DR's : $2, -2, -1$;

$$CD : \frac{-3\lambda + 7 - 2\mu - 5}{2} = \frac{2\lambda + 6 - \mu - 3}{-2} = \frac{4\lambda + 2 - 3\mu - 4}{-1}$$

$$CD : \frac{-3\lambda - 2\mu + 2}{2} = \frac{2\lambda - \mu + 3}{-2} = \frac{4\lambda - 3\mu - 2}{-1}$$

$$\lambda = 2, \quad \mu = 1$$

$$\therefore C : (1, 10, 10) ; D (7, 4, 7)$$

$$CD = \sqrt{6^2 + 6^2 + 3^2}$$

$$= \sqrt{36 + 36 + 9} = \sqrt{81} = 9 \text{ Ans.}$$

V01/3/079. The equation of plane parallel to line L_1 and containing line L_2 is equal to :

(A) $x + 3y + 4z = 30$

(B) $x + 2y + z = 15$

(C) $2x - y + z = 11$

(D) $2x + 17y - 7z = 33$

Ans. D

Sol. Equation of plane parallel to L_1 and containing L_2 is

$$a(x - 5) + b(y - 3) + c(z - 4) = 0$$

$$\therefore 2a + b + 3c = 0$$

$$\text{and } -3 + 2b + 4c = 0$$

$$\Rightarrow \frac{a}{-2} = \frac{b}{-17} = \frac{c}{7}$$

$$\text{So required plane : } 2x + 17y - 7z = 33$$

V01/3/080. The volume of parallelopiped formed by $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{AD}$ is equal to :

(A) 140

(B) 138

(C) 134

(D) 130

Ans. B

Sol. $A(7, 6, 2)$

$B(5, 3, 4)$

$C(1, 10, 10) D(7, 4, 7)$

$$\overrightarrow{AB} = -2\hat{i} - 3\hat{j} + 2\hat{k}$$

$$\overrightarrow{AC} = -6\hat{i} + 4\hat{j} + 8\hat{k}$$

$$\overrightarrow{AD} = 0\hat{i} - 2\hat{j} + 5\hat{k}$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & -3 & 2 \\ -6 & 4 & 8 \end{vmatrix}$$

$$= \hat{i}(-24 - 8) - \hat{j}(-16 + 12) + \hat{k}(-8 - 18)$$

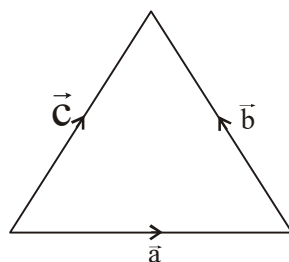
$$= -32\hat{i} + 4\hat{j} - 26\hat{k}$$

$$(\overrightarrow{AB} \times \overrightarrow{AC}) \cdot \overrightarrow{AD} = 0 - 8 - 13 = -138$$

$$\therefore \text{Volume} = \left| (\overrightarrow{AB} \times \overrightarrow{AC}) \cdot \overrightarrow{AD} \right| = |-138| = 138$$

INTEGER TYPE QUESTIONS

V01/4/081. If $\vec{a} = (\lambda x)\hat{i} + (y)\hat{j} + (4z)\hat{k}$, $\vec{b} = y\hat{i} + x\hat{j} + 3y\hat{k}$, $\vec{c} = -z\hat{i} - 2z\hat{j} - ((\lambda + 1)x)\hat{k}$ are sides of triangle as shown in figure then value of λ is (where x, y, z are not all zero) _____.



Ans. 0

Sol. $\vec{a} + \vec{b} = \vec{c}$

$$((\lambda x)\hat{i} + y\hat{j} + 4z\hat{k}) + (y\hat{i} + x\hat{j} + 3y\hat{k})$$

$$= -z\hat{i} - 2z\hat{j} - (\lambda + 1)x\hat{k}$$

$$\Rightarrow \lambda x + y + z = 0; \quad x + y + 2z = 0$$

$$\& (\lambda + 1)x + 3y + 4z = 0$$

$$\begin{vmatrix} \lambda & 1 & 1 \\ 1 & 1 & 2 \\ (\lambda + 1) & 3 & 4 \end{vmatrix} = 0$$

$$\lambda(4 - 6) - (4 - 2(\lambda + 1)) + (3 - (\lambda + 1)) = 0$$

$$-2\lambda - 4 + 2\lambda + 2 + 3 - \lambda - 1 = 0$$

$$-\lambda = 0 \Rightarrow \lambda = 0$$

V01/4/082. A sphere is inscribed in a tetrahedron whose vertices are $A \equiv (6, 0, 0)$, $B \equiv (0, 4, 0)$, $C \equiv (0, 0, 2)$ and $D \equiv (0, 0, 0)$. If the radius of sphere is $\frac{m}{n}$ (where m and n are relatively prime), then $(m + n)$ is _____.

Ans. 5

Sol. If S_1, S_2, S_3, S_4 are areas of four faces of tetrahedron

$$\Rightarrow \frac{1}{3}r(S_1 + S_2 + S_3 + S_4) = V$$

(where V is volume of tetrahedron)

$$\therefore r = \frac{3V}{S_1 + S_2 + S_3 + S_4}$$

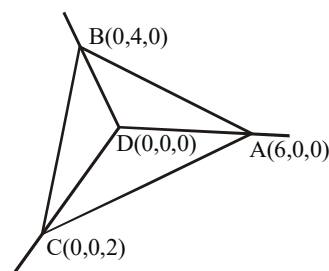
Now,

$$\therefore S_1 = 12, S_2 = 4,$$

$$S_3 = 6, S_4 = 14 \text{ and } V = \frac{1}{6} \times 48 = 8$$

$$r = \frac{24}{36} = \frac{2}{3}$$

$$\therefore m + n = 5$$



V01/4/083. Locus of all the points which are at a distance of 3 units from the line $\vec{r} = \lambda(\hat{i} + \hat{j} + \hat{k})$ is given by

$$x^2 + y^2 + z^2 - xy - yz - zx = \frac{k}{2} \text{ then the value of } \left(\frac{k}{3}\right) \text{ is equal to } \underline{\hspace{2cm}}.$$

Ans. 9

Sol. $\overline{QP} \cdot (\hat{i} + \hat{j} + \hat{k}) = 0$

$$\Rightarrow x - \lambda + y - \lambda + z - \lambda = 0$$

$$\Rightarrow \lambda = \frac{x + y + z}{3}$$

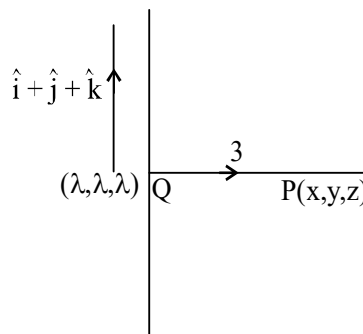
$$\text{Now } PQ^2 = 9$$

$$\Rightarrow (x - \lambda)^2 + (y - \lambda)^2 + (z - \lambda)^2 = 9$$

$$\text{Put } \lambda = \frac{x + y + z}{3} \text{ to get}$$

$$x^2 + y^2 + z^2 - xy - yz - zx = 27/2$$

$$\therefore k = 27 \Rightarrow k/3 = 9$$



V01/4/084. If α, β, γ & x, y, z are non-zero real numbers such that $(\alpha x + \beta y + \gamma z) + \sqrt{(\alpha^2 + \beta^2 + \gamma^2)(x^2 + y^2 + z^2)} = 0$,

$$\text{then the value of } \sqrt{\frac{\alpha y}{x \beta}} + \sqrt{\frac{\beta z}{y \gamma}} \text{ is equal to } \underline{\hspace{2cm}}.$$

Ans. 2

Sol. Let $\vec{p} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ and $\vec{q} = x\hat{i} + y\hat{j} + z\hat{k}$

$\therefore$ Given condition

$$\Rightarrow \vec{p} \cdot \vec{q} = -|\vec{p}||\vec{q}| \Rightarrow \cos \theta = -1 \Rightarrow \theta = \pi$$

$\therefore$ vector $\vec{p}$ and $\vec{q}$ are antiparallel

$$\therefore \frac{\alpha}{x} = \frac{\beta}{y} = \frac{\gamma}{z} = k \text{ (say)}$$

$$\Rightarrow \alpha = kx, \beta = ky, \gamma = kz$$

$$\text{then } \sqrt{\frac{\alpha y}{x\beta}} + \sqrt{\frac{\beta z}{y\gamma}} = 1 + 1 = 2.$$

V01/4/085. Let A, B, C be three vertices of a triangle. If $\vec{u} = \frac{1}{2}\vec{CA} + \frac{1}{3}\vec{CB}$ and X be a point on straight line passing through A and B such that $\vec{CX}$ is parallel to $\vec{u}$. If $\vec{CX} = \lambda\vec{CA} + \mu\vec{CB}$, then the value of $\frac{\lambda}{\mu}$ is _____.

Ans. 1.50

Sol. Let 'C' be the origin and $\vec{CA} = \vec{a}$ and $\vec{CB} = \vec{b}$ be the position vectors of A and B

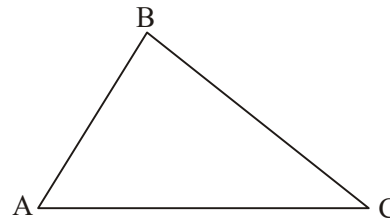
$$\vec{x} = \vec{a} + t(\vec{a} - \vec{b}), t \in \mathbb{R}$$

$$\vec{a} + t(\vec{a} - \vec{b}) = k\left(\frac{1}{2}\vec{a} + \frac{1}{3}\vec{b}\right) \quad \{\because \vec{CX} \parallel \vec{u}\}$$

$$1 + t = \frac{k}{2}; -t = \frac{k}{3}$$

$$1 - \frac{k}{3} = \frac{k}{2} \Rightarrow 1 = \frac{5k}{6} \Rightarrow k = \frac{6}{5}$$

$$\Rightarrow \vec{CX} = \frac{3}{5}\vec{a} + \frac{2}{5}\vec{b} \Rightarrow \frac{\lambda}{\mu} = \frac{3}{2}$$



V01/4/086. Three vectors $\vec{\alpha}, \vec{\beta}, \vec{\gamma}$ having magnitude 2, 3 and $5^{1/3}$ respectively satisfying $\vec{\gamma} \times ((\vec{\alpha} \times \vec{\gamma}) \times \vec{\gamma}) = \vec{\beta}$, then $\sin(\vec{\alpha} \wedge \vec{\gamma})$ is equal to _____.

Ans. 0.30

Sol. $|\alpha| = 2, |\beta| = 3, |\gamma| = 5^{\frac{1}{3}}$

$$\vec{\gamma} \times ((\vec{\alpha} \times \vec{\gamma}) \times \vec{\gamma}) = \vec{\beta}$$

$\vec{\gamma} \perp (\vec{\alpha} \times \vec{\gamma}) \times \vec{\gamma}$, take magnitude

$$5^{1/3} |(\alpha \times \gamma) \times \gamma| = 3$$

$$(\vec{\alpha} \times \vec{\gamma}) \perp \gamma$$

$$|\alpha \times \gamma| \cdot 5^{2/3} = 3$$

$$2 \cdot 5^{2/3} \cdot 5^{1/3} \cdot \sin(\vec{\alpha} \wedge \vec{\gamma}) = 3$$

$$\sin(\vec{\alpha} \wedge \vec{\gamma}) = \frac{3}{10}$$

V01/4/087. Let L_1 be the projection line of a line $L : \frac{x-1}{4} = \frac{y+1}{1} = \frac{z}{3}$ on the plane $\Pi : 3x - 4z - 2 = 0$. P and Q are the points on L_1 corresponding to $A \equiv (1, -1, 0)$ and $B \equiv (5, 0, 3)$ on the line L. If R is the point on the line L which divides AB externally in the ratio 31 : 5 then area of $\Delta PQR = \sqrt{\frac{p}{q}}$ where p, q $\in \mathbb{N}$. Find the least value of $|4p - q|$ _____.

Ans. 2

Sol. Line L is parallel to the plane.

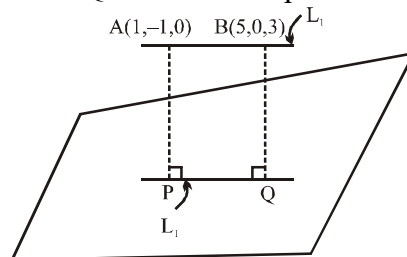
L_1 is the projection line on the plane and P and Q are foot of the parallel on the plane.

R is a point on the line L

$$\text{Area of } \Delta PQR = \frac{1}{2} \times PQ \times AP$$

$$= \frac{1}{2} \times \sqrt{26} \times \frac{1}{5} = \sqrt{\frac{26}{100}} = \sqrt{\frac{13}{50}}$$

$$\therefore 4p - q = 52 - 50 = 2$$



V01/4/088. Number of points having positive integral co-ordinate lying on the plane $x + 2y + 3z = 15$ is n, then

$\frac{n}{2}$ is equal to _____.

Ans. 6

Sol.	$x + 2y + 3z = 15$			
	Put $z = 1$	$z = 2$	$z = 3$	$z = 4$
	$x + 2y = 12$	$x + 2y = 9$	$x + 2y = 6$	$x + 2y = 3$
	(10,1,1)	$x = 7, y = 1$	$x = 4, y = 1$	(1,1,4)
	(8,2,1)	(7,1,2)	$x = 2, y = 2$	
	(6,3,1)	(5,2,2)	(4,1,3)	
	(4,4,1)	(3,3,2)	(2,2,3)	
	(2,5,1)	(1,4,2)		
	$n = 12$			

V01/4/089. Let P be a plane containing the line $z = 2, x = -y$ and passing through a point $M(\alpha, \beta, \gamma)$ where $\alpha > 0, \beta > 0, \gamma > 0$. The distance of M from yz plane, zx plane and xy plane are 1, 2 and 3 respectively. Plane P meets co-ordinate axes at points A, B and C respectively. Find the volume of tetrahedron $OABC$ (O is the origin). _____.

Ans. 12

Sol.
$$\frac{x-1}{1} = \frac{y+1}{-1} = \frac{z-2}{0}$$

$P(1, -1, 2), M(1, 2, 3)$

DR's of $AB : (0, -3, -1)$

$$\hat{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -3 & -1 \\ 1 & -1 & 0 \end{vmatrix} = -\hat{i} - \hat{j} + 3\hat{k}$$

$\langle -1, -1, 3 \rangle$

$-(x-1) - (y-2) + 3(z-3) = 0$

$-x - y + 3z + 1 + 2 - 9 = 0$

$x + y - 3z = -6$

pts $(-6, 0, 0), (0, -6, 0)$ and $(0, 0, 2)$

Volume $= \frac{1}{6} [\overrightarrow{OA} \overrightarrow{OB} \overrightarrow{OC}]$

$$= \frac{1}{6} \begin{vmatrix} -6 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & 2 \end{vmatrix}$$

$= \frac{1}{6} \times 6 \times 12 = 12$

V01/4/090. In a regular tetrahedron, the centres of the four faces are the vertices of a smaller tetrahedron. The ratio of the volume of the smaller tetrahedron to that of the larger is $\frac{m}{n}$, where m and n are relatively prime positive integers. Find the value of (m + n) _____.

Ans. 28

Sol. Let vertices be (1, 0, 0, 0), (0, 1, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1)

The side of length of larger = $\sqrt{2}$, the side length of small = $\frac{\sqrt{2}}{3}$.

$\therefore$ volume $\propto$ side³

$$\text{Ratio of volume} = \frac{\left(\frac{\sqrt{2}}{3}\right)^3}{(\sqrt{2})^3} = \frac{1}{27} = \frac{m}{n}$$

$$m + n = 28$$

V01/4/091. Let (p_1, q_1, r_1) & (p_2, q_2, r_2) are satisfying $\begin{vmatrix} 1 & p & p^2 \\ 1 & q & q^2 \\ 1 & r & r^2 \end{vmatrix} = 6$ (where $p_i, q_i, r_i \in \mathbb{N}$ and $p_i < q_i < r_i$ & $i = 1, 2$) and

point (p_1, q_1, r_1) lies on the plane $2x + 3y + 6z = k_1$ and point (p_2, q_2, r_2) lies on the plane $2x + 3y + 6z = k_2$ (where $p_1 = p_2 = 1$) If distance between these planes is 'd', then value of (210d) is _____.

Ans. 90

Sol. $(p - q)(q - r)(r - p) = 6$

$$\therefore p - q = -1, q - r = -2, r - p = 3$$

$$\text{or } p - q = -2, q - r = -1, r - p = 3$$

$$(p_1, q_1, r_1) = (1, 2, 4)$$

$$(p_2, q_2, r_2) = (1, 3, 4)$$

$$d = \frac{|k_1 - k_2|}{7} = \frac{3}{7}$$

$$210d = 90$$

V01/4/092. Let $\pi : x + y - z - 4 = 0$ be the equation of a plane and A be the point with position vector $\hat{i} + 2\hat{j} - 3\hat{k}$. L is a line which passes through the point (1, 2, 3) with direction ratios 3, -1 and 4. If the distance of the point A from the line L measured parallel to the plane π is d_1 and the distance of the point A from the plane π measured parallel to the line L is d_2 , then find the value of $\sqrt{d_1^2 - d_2^2}$ _____.

Ans. 10

Sol. Plane $\pi : x + y - z = 0$

$$\overrightarrow{OA} = \hat{i} + 2\hat{j} - 3\hat{k}$$

$$\text{Eqn. L : } \frac{x-1}{3} = \frac{y-2}{-1} = \frac{z-3}{4} = \lambda \text{ (let)}$$

Point P(3λ + 1, -λ + 2, 4λ + 3) lies on the line.

Point A(1, 2, -3)

DRs. of line from A and P

$$(3\lambda, -\lambda, 4\lambda - 6)$$

$$3\lambda - \lambda - 4\lambda - 6 = 0$$

$$-2\lambda - 6 = 0$$

$$\lambda = -3$$

$$AB = d_1 = \sqrt{9\lambda^2 + \lambda^2 + 16\lambda^2 + 36 + 48\lambda}$$

$$d_1 = \sqrt{126}$$

$$\Rightarrow \text{equ. of || line } \frac{x-1}{3} = \frac{y-2}{-1} = \frac{z+3}{4} = \lambda$$

$$x = 3\lambda + 1, y = 2 - \lambda, z = 4\lambda - 3$$

$$AC : (3\lambda, -\lambda, 4\lambda)$$

$$3\lambda + 1 - \lambda + 2 - (4\lambda - 3) = 4$$

$$2\lambda - 4\lambda + 6 = 4$$

$$-2\lambda = -2$$

$$\lambda = 1$$

$$AC = \sqrt{(9+1+16)\lambda^2}$$

$$d_2 = \sqrt{26 \times 1} = \sqrt{26}$$

$$\sqrt{d_1^2 - d_2^2} = \sqrt{126 - 26} = 10$$

V01/4/093. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $|\vec{a}| = 1$ and $\vec{a} \cdot (\vec{b} \times (\vec{a} \times \vec{b})) = 8$. If the angle between $\vec{a}$ and $\vec{b}$ is $\text{cosec}^{-1} \sqrt{2}$, then magnitude of $\vec{b}$ is _____.

Ans. 4

Sol. $|\vec{a}| = 1$

$$Q = \text{cosec}^{-1} \sqrt{2}$$

$$Q = \frac{\pi}{4}$$

$$\therefore \vec{a} \cdot (\vec{b} \times (\vec{a} \times \vec{b})) = 8$$

$$\vec{a} \cdot (\vec{b} \times |\vec{a}| \cdot |\vec{b}| \sin \frac{\pi}{4}) = 8$$

$$\vec{a} \cdot (\vec{b} \cdot 1 \cdot |\vec{b}| \cdot \frac{1}{\sqrt{2}}) = 8$$

$$|\vec{a}| \cdot |\vec{b}| \cdot \cos \frac{\pi}{4} \cdot |\vec{b}| = 8\sqrt{2}$$

$$1 \cdot |\vec{b}|^2 \times \frac{1}{\sqrt{2}} = 8\sqrt{2}$$

$$|\vec{b}| = 4$$

V01/4/094. Let P_1 be a plane passing through two points $A(1, 2, 0)$ and $B(2, -1, 1)$ and perpendicular to $\vec{r} \cdot (\hat{i} + \hat{j} - 3\hat{k}) = 1$. Let L be a line through $(1, 1, 1)$ and perpendicular to line of intersection of planes P_1 and P_2 where equation of plane P_2 is $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 5$. If the equation of line L is $\frac{x+1}{-1} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$, then find the value of

$$\left(\frac{2y_1 + z_1}{bc} \right) \text{ —————.}$$

Ans. 9

Sol.
$$\vec{x} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -3 & 1 \\ 1 & 1 & -3 \end{vmatrix} = 8\hat{i} + 4\hat{j} + 4\hat{k}$$

$$\hat{n} = \langle 2, 1, 1 \rangle$$

$$2x + y + z = 4 \rightarrow P_1$$

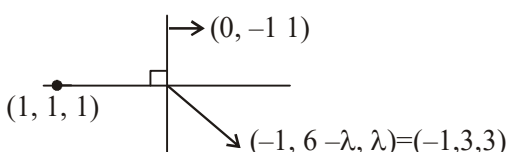
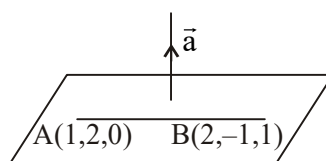
$$x + y + z = 5 \rightarrow P_2$$

$$\vec{P}_1 \times \vec{P}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = \langle 0, -1, 1 \rangle$$

$$\text{Let } z = 0 \Rightarrow \text{P. O. I} = (-1, 6, 0)$$

$$\frac{x+1}{0} = \frac{y-6}{-1} = \frac{z-0}{1} = \lambda$$

$$\langle -2, 5 - \lambda, \lambda - 1 \rangle \cdot \langle 0, -1, 1 \rangle = 0$$



$$-5 + \lambda + \lambda - 1 = 0$$

$$\Rightarrow \lambda = 3$$

$$L: \langle -2, 2, 2 \rangle = \langle -1, 1, 1 \rangle$$

$$(1, 1, 1) \quad \quad \quad (-1, 3, 3)$$

$$\frac{x+1}{-1} = \frac{y-3}{1} = \frac{z-3}{1}$$

$$y_1 = 3, z_1 = 3, b = c = 1$$

$$\frac{2y_1 + z_1}{bc} = \frac{2 \times 3 + 3}{1} = 9$$

V01/4/095. The radical centre of the following set of circles $x^2 + y^2 - 3x - 6y + 14 = 0$; $x^2 + y^2 - x - 4y + 8 = 0$; $x^2 + y^2 + 2x - 6y + 9 = 0$ is (α, β) . Find the equations of the straight line passing through the point $(\alpha, \beta, 3)$ to intersect the straight line $x + 1 = 2(y - 2) = z + 4$ and parallel to the plane $x + 5y + 4z = 0$ _____.

Ans. $\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-3}$

Sol. $S_1 - S_2 = 0$
 $2x + 2y = 6$
 $x + y = 3 \quad \quad \quad \dots(1)$
 $S_2 - S_3 = 0$
 $3x - 2y + 1 = 0 \quad \quad \quad \dots(2)$
 From eq. (1) & (2)
 Radical center $(1, 2)$

Point $P(1, 2, 3)$ & Line $\frac{x+1}{2} = \frac{y-2}{1} = \frac{z+4}{2} = \lambda$

$Q(2\lambda - 1, \lambda + 1, 2\lambda - 4)$

$\overrightarrow{PQ} \cdot \vec{n} = 0$

$1(2\lambda - 2) + 5\lambda + 4(2\lambda - 7) = 0$

$15\lambda = 30$

$\lambda = 2$

$Q(3, 4, 0) \Rightarrow$ Line $\frac{x-1}{2} = \frac{y-2}{2} = \frac{z-3}{-3}$

V01/4/096. Let M and M' be the feet of perpendiculars from foci S and S' to the tangent at any point P on an ellipse. If $SM = 2S'M'$ and $S'P = 4$ then SP is equal to λ . Feet of the perpendicular drawn from the point P $(\lambda, 3, -5)$ on the axes of coordinates are A, B and C. Find the equation of the plane passing through their feet and the area of ΔABC _____.

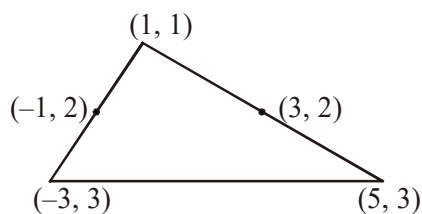
Ans. $\frac{x}{2} + \frac{y}{3} + \frac{z}{-5} = 1$, Area = $\frac{19}{2}$ sq. units

Sol.

V01/4/097. If the vertex of a triangle is (1, 1) and the midpoint of two sides of a triangle through this vertex are (-1, 2) and (3, 2) then the centroid of the triangle is $\left(1, \frac{\alpha}{\beta}\right)$. If a, b, c $\in \mathbb{N}$, then number of points having position vector $a\hat{i} + b\hat{j} + c\hat{k}$ such that $5 \leq a + b + c \leq (\alpha + \beta)$ is _____.

Ans. 116

Sol.



Centroid $\left(\frac{1-3+5}{3}, \frac{1+3+3}{3}\right)$

$\left(1, \frac{7}{3}\right)$

$\alpha = 7 \quad \beta = 3$

$\alpha + \beta = 10$

$5 \leq a + b + c \leq 10$

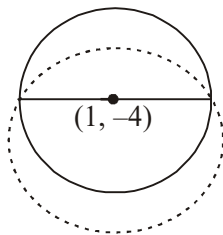
$a + b + c = 5, 6, 7, 8, 9, 10$

${}^4C_2 + {}^5C_2 + {}^6C_2 + {}^7C_2 + {}^8C_2 + {}^9C_2$
 $= {}^{10}C_3 - {}^4C_3 = 120 - 4 = 116$

V01/4/098. If the circle $x^2 + y^2 + 4x + 22y + a = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - b = 0$ (where a, b > 0), then the maximum value of (ab) is λ . The plane denoted by $\Pi_1 : 4x + 7y + 4z + 81 = 0$ is rotated through a right angle about its line of intersection with the plane $\Pi_2 : 5x + 3y + 10z = \sqrt{\lambda}$. If the plane in its new position be denoted by Π , and the distance of this plane from the origin is $\sqrt{k}$ where $k \in \mathbb{N}$, then find k _____.

Ans. 212

Sol.



$$S_1 - S_2 = 0$$

$$6x + 14y + a + b = 0$$

$$(1, -4)$$

$$6 - 56 + a + b = 0$$

$$a + b = 50$$

$$AM \geq GM$$

$$\frac{a+b}{2} \geq \sqrt{ab}$$

$$ab \leq 25^2$$

$$ab \leq 625$$

$$\lambda = 625$$

$$\text{Let new plane be } P_1 + \lambda P_2 = 0$$

$$(4x + 7y + 4z + 81) + \lambda(5x + 3y + 10z - 25) = 0$$

$$(5\lambda + 4)x + (3\lambda + 7)y + (10\lambda + 4)z + 81 - 25\lambda = 0$$

$$\text{perpendicular to } P_1$$

$$\Rightarrow 4(5\lambda + 4) + 7(3\lambda + 7) + 4(10\lambda + 4) = 0$$

$$81\lambda = -81$$

$$\lambda = -1$$

$$\text{New plane be } -x + 4y - 6z + 106 = 0$$

$$(0,0,0)$$

$$P = \frac{106}{\sqrt{1+16+36}} = \sqrt{\frac{106 \times 106}{53}} = \sqrt{212}$$

V01/4/099. A straight line L intersects perpendicularly both the lines $\frac{x+2}{2} = \frac{y+6}{3} = \frac{z-34}{-10}$ and

$$\frac{x+6}{4} = \frac{y-7}{-3} = \frac{z-7}{-2}, \text{ then the square of perpendicular distance of origin from L is } \underline{\hspace{2cm}}.$$

Ans. 5

Sol. Line perpendicular to both lines will have direction ratios (2, 2, 1).

$$\therefore \begin{bmatrix} i & j & k \\ 2 & 3 & -10 \\ 4 & -3 & -2 \end{bmatrix}$$

$$\therefore A(-2 + 2r, -6 + 3r, 34 - 10r)$$

$$B(-6 + 4\ell, 7 - 3\ell, 7 - 2\ell)$$

$$\therefore \frac{4 + 2r - 4\ell}{2} = \frac{3r + 3\ell - 13}{2} = \frac{2\ell - 10r + 27}{1}$$

$$r = 3 \text{ and } \ell = 2$$

$$\therefore L_3: \frac{x-4}{2} = \frac{y-3}{2} = \frac{z-4}{1}$$

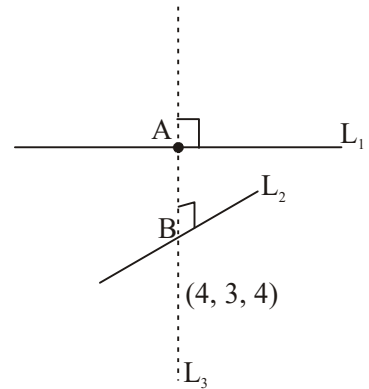
feet of $\perp$ from origin to $L_3(4 + 2t, 3 + 2t, 4 + t)$

$$\therefore (4 + 2t) \cdot 2 + (3 + 2t) \cdot 2 + (4 + t) \cdot 1 = 0$$

$$\Rightarrow t = -2 \quad \Rightarrow (0, -1, 2)$$

$\therefore \perp$ distance from $(0, 0, 0)$ to L_3

$$= \sqrt{0^2 + (1)^2 + (-2)^2}$$



V01/4/100. If $\hat{a}, \hat{b}$ and $\hat{c}$ are non-coplanar unit vectors such that $[\hat{a} \hat{b} \hat{c}] = [\hat{b} \times \hat{c}, \hat{c} \times \hat{a}, \hat{a} \times \hat{b}]$ then find the projection of $\hat{b} + \hat{c}$ on $\hat{a} \times \hat{b}$. _____.

Ans. 1

Sol.
$$[\hat{a} \hat{b} \hat{c}] = [\hat{b} \times \hat{c} \quad \hat{c} \times \hat{a} \quad \hat{a} \times \hat{b}] = [\hat{a} \hat{b} \hat{c}]^2$$

$$\Rightarrow [\hat{a} \hat{b} \hat{c}] = 1$$

$$\text{projection of } \hat{b} + \hat{c} \text{ on } \hat{a} \times \hat{b} = \frac{(\hat{b} + \hat{c}) \cdot (\hat{a} \times \hat{b})}{|\hat{a} \times \hat{b}|} = \frac{[\hat{a} \hat{b} \hat{c}]}{|\hat{a} \times \hat{b}|}$$